

GOVERNMENT OF KARNATAKA



DEPARTMENT OF TECHNICAL AND COLLEGIATE EDUCATION

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING



5th Semester Pathway Specialization

RENEWABLE ENERGY

(20EE53I)

Name of the student :

Register number :

Unit 1

- 1. Consider a 500-watt PV array that operates at 24 V DC and has a battery bank of 12 V DC. Determine an MPPT charge controller rating for this given system.**

Solar PV array wattage = 500 W

Solar PV arrays operating voltage = 24 V

Battery pack operating voltage = 12 V

Solution:

(i) The input power to the MPPT controller is 500 W, the solar PV array is connected at the input side of the MPPT charge controller and the battery is connected at the output side of the MPPT charge controller. Thus, the battery acts as a load to the system. The data specifies the output voltage. Assuming 100 % efficiency we can determine the output current for its load.

$$\text{Power} = \text{Voltage} \times \text{Current}$$

$$\text{Current} = \text{Power} / \text{Voltage} = 500\text{W} / 12\text{V} = 41.66 \text{ A}$$

Thus, we would require a 12 V, 41.66 A MPPT for the above system, we can increase the current value by 25 % considering some conditions that occur causing the panel to produce more power. Thus, we can take it as 52 A. So, 12 V, 52 A MPPT charge controller would be suitable for the above system. Note that the MPPT charge controller should be able to handle the open-circuit voltage and the voltage at the maximum power point of the connected PV array.

- 2. Compare merits and demerits of solar photovoltaic system.**

Merits of solar photovoltaic system:

- Solar energy is freely available in nature
- Direct conversion of sun light into electricity
- Absence of moving parts
- Low maintenance cost
- No environmental pollution
- Long life and highly reliable
- It can be easily started as no starting time is required
- It can be installed at load centers and hence saves cost of transmission and Distribution.

Demerits of solar photovoltaic system:

- Solar power plants require large area of land
- It needs high capital cost
- Solar energy is unreliable and therefore storage batteries are needed
- Manufacturing of solar array requires heavy labour

3. **A 40W TV runs for 10 hours and a 70W fan runs for 5 hours. Considering the DOD=80%, Discharge Efficiency=85% and the battery voltage as 12V, what should be the ideal capacity of the battery?**

$$\begin{aligned}\text{Total Load in Wh} &= (40 \times 10) + (70 \times 5) \\ &= 750 \text{ Wh}\end{aligned}$$

$$\text{Battery capacity} = \frac{\text{Total energy required per day} \times \text{No. of days}}{\text{Depth of discharge} \times \text{Discharge efficiency} \times \text{nominal voltage of the battery}}$$

$$\begin{aligned}\text{Battery capacity required} &= (750 \times 3) / (0.8 \times 0.85 \times 12) \\ &= 275.73 \text{ Ah}\end{aligned}$$

From the batteries available in market, we choose 2 batteries of 180Ah, 12V connected in parallel.

4. **What is the meaning of inverter, Differentiate between the terms Central inverters, String inverters, Micro inverters?**

Inverter: The inverter is the equipment that converts direct current to alternating current and controls the quality of the output power to be delivered to the grid.

Central inverter: Central inverters are used for large grid-connected solar systems. A central inverter may have a single MPPT or multiple MPPTs. Most of the central inverters in the solar market today have only one MPPT. PV array is divided into a number of sub-arrays comprising of a number of strings.

String inverters; String inverters range from 1 kW up to approximately 35 kW, and are connected to one or multiple strings of a PV array to a single maximum power point tracking

(MPPT). A “string” is made of PV modules connected in series. Multiple inverters can be used to spread the generated power over more than one phase.

Micro inverters: Micro inverters or modular inverters are small inverters that are designed to be mounted on the back of the solar module. The capacity of micro inverter is equal to the module capacity to which it is connected.

5. (a) Discuss the terms Photovoltaic (PV) cell, Module, Array

Photovoltaic (PV): The energy conversion device that converts sunlight into electricity by the use of photovoltaic effect is called as solar cell. A solar cell is also known as photovoltaic cell or photo electric cell.

Solar photovoltaic module: The output voltage of a single cell is very small and is about 0.3 to 0.5V. To increase the voltage level, several cells are interconnected to form a photovoltaic module. The number of cells in a module depends on the module voltage. A 12V solar module consists of 33 to 36 cells. These cells are mounted on a durable back cover and are sealed with a transparent top cover to protect against dust, moisture, mechanical shocks and other outdoor conditions.

Solar photovoltaic array: Several solar modules are interconnected in series and parallel to form solar arrays. These modules are connected in series to increase the voltage and they are connected in parallel to increase the current output. There are two types of solar arrays namely

- Tracking arrays
- Fixed arrays

Tracking arrays are movable type and they are always aligned perpendicular to the direction of sun rays to receive maximum solar radiation.

A fixed array is placed east-west and tilted at an angle equal to the latitude of that place.

6. **Discuss the necessity of charge controller, design a 140 WP Solar home system with a PV module of 70 W having an open-circuit voltage of 20 V and voltage at maximum power point of 16 V. The voltage of the battery bank is at 12 V. Determine a suitable MPPT charge controller rating for this home solar design.**

A battery charge controller (BCC) regulates the flow of electricity from the PV generator to the battery. Its function is to regulate the voltage and current from the PV array in order to prevent overcharging and also over discharging of the battery

Let us connect the available PV module in series.

Thus, the open-circuit voltage of the system would become = $2 \times 20 = 40 \text{ V}$

Voltage at maximum power point would be = $2 \times 16\text{V} = 32 \text{ V}$

Peak power of the system would be = $2 \times 70\text{W} = 140 \text{ W}$

The input power to the MPPT charge controller is 140 W if we assume 100 % efficiency.

And the available battery voltage at 12 V, then the current to the battery pack can be determined as follows;

Power = Voltage $\times$ Current

Current = Power / Voltage = $140\text{W} / 12\text{V} = 11.66 \text{ A}$

Thus, **we would require a 12 V, 11.66 A MPPT** for the above system, we can increase the current value by 25 % considering some conditions that occur causing the panel to produce more power. Thus, we can take it as 15 A. **So, 12 V, 15 A MPPT charge controller would be suitable** for the above system.

Again, it is important to note that the MPPT charge controller should be able to handle the open-circuit voltage and the voltage at the maximum power point of the connected PV array.

7. **Calculate the peak capacity of the panel and the energy generated considering the effective sunshine hours to be 5 hours for given solar panel specification**

V_{oc}	43.42V
I_{sc}	4.48A
V_{mp}	35.82V
I_{mp}	4.26A

Solar pane max power: $V_{mp} \times I_{mp} = 35.82 \times 4.2 = 150 \text{ Wp}$

Peak capacity of the solar panel at 5 hours' effective sunshine hours is $150 \times 5 = 750 \text{ watts}$.

8. Calculate the battery capacity needed for a 20w LED tube-light that is used for 10 hours. Consider the battery rating to be 12V.

$$\text{Total Energy Demand} = 20\text{w} \times 10\text{h} = 200 \text{ wh}$$

$$\text{Battery Size} = \frac{\text{Total Energy Demand} \times \text{Days of Autonomy}}{\text{Battery Efficiency} \times \text{DOD} \times \text{Nominal Voltage}}$$

$$\text{Battery Size} = \frac{200 \text{ wh} \times 2}{0.85 \times 0.6 \times 12 \text{ V}}$$

$$\text{Battery capacity} = 66 \text{ Ah}$$

9. Discuss the relation between GHI, DNI and DHI with their definition.

Direct Normal Irradiance (DNI) is the amount of solar radiation received directly when the panel is perpendicular to sun.

DNI is acronym for Diffused Normal Irradiance which represents the amount of light that is coming perpendicular to surface. The surface here represents ground or something parallel to ground.

Diffuse horizontal radiation (DHI) is the component of the sunlight reaching the panels that is reflected by clouds and other objects

(OR)

Diffuse Horizontal Irradiance is the amount of radiation received per unit area by a surface that does not arrive on a direct path from the sun, but has been scattered by molecules and particles in the atmosphere. Basically, it is the illumination that comes from clouds and the blue sky

➤ **Relation between GHI, DNI and DHI.**

If GHI cannot be measured directly, it may be calculated from direct normal irradiance (DNI) and diffuse horizontal irradiance (DHI) using the following equation:

$$\text{GHI} = \text{DHI} + \text{DNI} \times \cos(\theta)$$

where θ is the solar zenith angle (vertically above the location is 0° , horizontal is 90°).

10. Customer wants to buy charge controller for his 3kw rooftop PV system. what type of charge controller and its rating you recommend the for a 24V battery. Justify your recommendation

For 3 kw rooftop PV system I recommend MPPT charge controller.

An MPPT solar charge controller stands for “Maximum Power Point Tracking”. It will measure the V_{mp} voltage of the panel, and down-converts the PV voltage to the battery voltage. Because power into the charge controller equals power out of the charge controller, when the voltage is dropped to match the battery bank, the current is raised, so you are using more of the available power from the panel. You can use a higher voltage solar array than battery, like the 60 cell nominal 20V grid-tie solar panels that are more readily available. With a 20V solar panel, you can charge a 12V battery bank, or two in series can charge up to a 24V battery bank, and three in series can charge up to a 48V battery bank. This opens up a whole wide range of solar panels that now can be used for your off-grid solar system.

Charge Controller is normally calculated by

- ✓ Battery bank Voltage
- ✓ Total solar panel power

$$\text{Charge Controller Current} = \frac{\text{Total solar panel power}}{\text{Battery bank Voltage}}$$

$$\text{Charge Controller Current} = \frac{3000}{24} = 125 \text{ A}$$

125 A would be the better rating of the charge controller for 3kw PV system.

11. Compare roof mounted and ground mounted solar PV system

Sl No	Roof mounted	Ground mounted
1	Installed directly on the roof	Installed in an open area or as a carport over a parking lot.
2	Panels can be attached to flat or sloping roofs consisting of metal, shingle, or rubber materials.	panels are secured to a rack structure that is connected to the ground with steel beams or another type of metal post.
3	Limits unauthorized visitors from accessing the panels	The system can easily be accessed for maintenance
4	The age and condition of your roof	The long-term plan for the plot of land
5	Typically the installation cost is lower	Typically the installation cost is higher

12. Write the advantages and disadvantages of solar photovoltaic system Lithium-ion batteries

Advantages

1. Lithium ion batteries require almost no regular maintenance.
2. They also have a higher battery energy density, meaning they can hold more energy in a smaller space than a lead acid battery.
3. Lithium ion batteries have a longer life cycle, or lifespan, as well - most have a guaranteed warranty of at least 10 years.

This longer lifespan has to do with lithium ion batteries having a higher depth of discharge, so you can use more of the energy stored within the battery before it has to be recharged.

Disadvantages

1. One of the biggest disadvantages of lithium ion batteries is that they are more expensive than other energy storage technologies.
2. Also, because of their chemistry, lithium ion storage systems have a higher chance of catching fire due to something called thermal runaway. However, if installed properly, the chance of your battery catching fire is close to zero.

13. Mr X has purchased a land in a remote village which has no access to electricity to build a energy efficient sustainable farm house. His requirements are as follows.

a. Highly efficient solar panel

b. Combine farming with PV

c. Buildings constructed using PV panels

d. Additional solar panel on the roof top which moves in the direction of the sun

e. What is this type of configuration called as ? With the help of a block diagram, explain its working

(a). Monocrystalline: - It has high efficiency compared to other Solar PV Cell

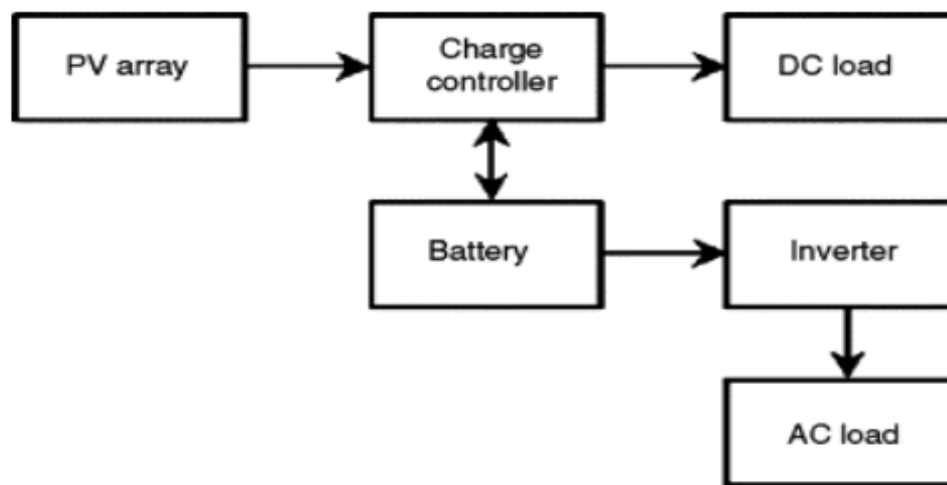
(b). Agrovoltatics: - Agrovoltatics combine energy generation and agriculture in the same land.

(c). Building Integrated Photo Voltaic (BIPV) system:- Here the building itself constructed by using Solar PV panels to generate electricity.

(d). Solar panel which moves in the direction of sun is called a solar tracker.

(e) Stand alone configuration or off grid system:- Since there is no access to electricity this system is best.

Standalone solar photovoltaic system



- It consists of photo voltaic array, charge regulator, battery, inverter, DC load centre and AC load centre.
- The DC output of the PV array is connected to the battery through the charge regulator unit.
- The charge regulator consists of a blocking diode in series with the photovoltaic array. It prevents the battery from being discharged through the PV array at night when there is no sunshine.
- The blocking diode also prevents overcharging and discharging of the battery and also protects the battery from short circuits.
- The battery gets charged from the DC output of the PV array.
- Battery output can be directly connected to the DC loads.
- For AC loads, the battery output is first connected to an inverter which converts DC to AC and then it is connected to the AC load centre.

14. Discuss between the terms Voc, Pmax, Vmp , Isc, Imp

Open Circuit Voltage (Voc): The open circuit voltage is the maximum voltage that the solar panel can produce with no load on it.

Maximum Power Point (Pmax): The Pmax is the combination of the volts and amps that results in the highest wattage (Volts x Amps = Watts).

Maximum Power Point Voltage (Vmp): The Vmp is the voltage generated when the power output is the greatest.

Short Circuit Current (Isc): Short Circuit Current is current the solar panels are producing when the plus and minus of the panels wires are directly connected to each other. This is the highest current the solar panels will produce under standard test conditions.

Imp Maximum Current: This is the current corresponding to maximum power.

15. What is the meaning of 100Ah battery? Calculate the battery capacity needed for a 20W LED tube-light that is used for 10 hours. Number of days of autonomy= 3, DOD=80%, Discharge Efficiency=85%. Consider the battery rating to be 12V.

100Ah Battery means the battery can supply a DC load of 100 Amps for duration of 1 Hour until the battery gets fully discharged or it can supply 2A Amps for 50 Hours or any other combination of load current requirement and the time duration

$$\begin{aligned}\text{Battery capacity} &= \frac{\text{Total energy required per day} \times \text{No. of days}}{\text{Depth of discharge} \times \text{Discharge efficiency} \times \text{nominal voltage of the battery}} \\ &= (20 \times 10 \times 3) / (0.8 \times 0.85 \times 12) \\ &= 73.5 \text{ Ah}\end{aligned}$$

From the batteries available in the market 100Ah 12V battery is chosen.

16. Compare the different types of charge controllers available. A 100Wp solar panel with nominal voltage 12V and Vmp=36V, is connected to a charge controller to charge a 12V battery. Analyse the efficiency of the different charge controllers.

The different types of charge controllers available are

- a. PWM charge controllers and
- b. MPPT charge controllers

PWM charge controllers: PWM stands for Pulse Width Modulation. Here the width of the voltage pulse coming from solar array is modulated to match it to battery voltage.

It is cheaper and less efficient compared to MPPT Charge controller.

MPPT charge controllers: MPPT stands for Maximum Power Point Tracking. It tracks the maximum power point and ensures that its output power is equal to its input power.

It is more efficient and hence costlier than PWM charge controller

In the above problem, $I_{mp} = 100/36 = 2.778 \text{ A}$

In case of PWM charge controllers, same current flows to charge the batteries. Therefore, its output = $2.778 \times 12 = 33.34 \text{ W}$, whereas its input is 100W.

Therefore, losses = $100 - 33.34 = 66.66 \text{ W}$

In case of MPPT charge controllers, input power = output power.

Therefore, $36 \times 2.778 = 12 \times \text{Battery charging current}$

Battery charging current = $(36 \times 2.778)/12 = 8.33 \text{ A}$

Therefore, the output of the MPPT charge controller = $8.33 \times 12 = 100\text{W}$. Hence there are no losses in case of MPPT charge controllers.

17. Differentiate between Solar Power Conditioning Unit (PCU) and solar Inverter

Solar Power Conditioning unit (PCU)

- Solar Power Conditioning unit (PCU) is an integrated system consisting of a solar charge controller, inverter and a Grid charger.
- It provides the facility to charge the battery bank through either a Solar or Grid/DG set.
- The PCU continuously monitors the state of battery voltage, solar power output and the load.
- Due to constant usage of power, if the battery voltage goes below a set level, the PCU will automatically transfer the load to the Grid/DG power and also charge simultaneously.
- The PCU always gives preference to the solar power and will use Grid/DG power only when the solar power / battery charger is unable to meet the load requirement

Solar Inverter

- Solar inverter has priority for solar charging similar to PCU. However, during solar day, **battery will be charged through solar but in presence of mains the load will be fed through mains through bypass mode.**
- Solar energy stored in battery can be utilized when AC mains is absent (in Inverter mode). This way solar energy will be stored in the battery and can be utilized when AC mains gets cutoff. **As long as the mains is present solar power is kept stored.**

18. A commercial building has the following loads 18 watts tube lights = 100 Nos, 60 watts fan = 50 Nos, 100 Watt Computers = 25 Nos and 1Kw AC = 4 Nos. Calculate the total load and select suitable size of solar inverter allowing suitable factor of safety.

Total Power = $18 \times 100 + 60 \times 50 + 100 \times 25 + 1000 \times 4 = 11300 \text{ W} = 11.3 \text{ Kw}$

Inverter sizing considering a factor of safety of 25% and p.f 0.8,

Inverter Size = $(\text{Total wattage} \times 1.25) / 0.8 = 17.65 \text{ KVA} \sim 20 \text{ KVA}$. Hence 20KVA, 240V inverter is chosen.

19. As a responsible citizen, Analyze how you can apply 6Rs of sustainability to your day to day life.

Rethink: Do we make too many products? Design in a way that considers people and the environment. **Refuse:** Don't use a material or buy a product if you don't need it or if it's bad for people or the environment.

Reduce: Cut down the amount of material and energy you use as much as you can.

Reuse: Use a product to make something else with all or parts of it.

Recycle: Reprocess a material or product and make something else.

Repair: When a product breaks down or doesn't work properly, fix it.

20. Difference between Net zero and Zero

1. The term net zero means achieving a balance between the carbon emitted into the atmosphere, and the carbon removed from it.
2. This balance – or net zero – will happen when the amount of carbon we add to the atmosphere is no more than the amount removed.
3. Net zero can also be defined as the way in which emissions are balanced by efforts to reduce them, which include “planting trees, changing industrial processes, restoring forests or using nature-friendly technology to store and capture carbon emitted from burning oil and coal.”
4. Net zero is all about ‘balancing’ or cancelling out any carbon we produce. We reach net zero when the amount of greenhouse gas we produce is no more than the amount taken away.
5. Zero carbon concerns the emissions produced from a product or service – it means no carbon is given off at all.

Unit 2

1. Mr. X wants to plan a project in a particular location, what are the different consideration he should take while doing the survey

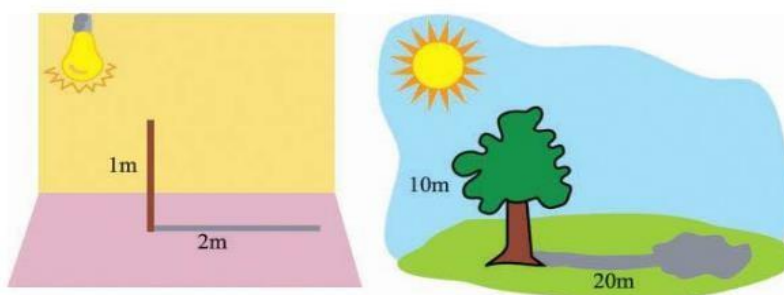
Identify location for installation PV array:

- The location of PV array should be selected such that it is –
- Free from shadow in all days of the year
- Has access for maintenance
- Provide ample space for air cooling
- Prevails aesthetic of the building or premises
- Not far from the inverter
- Protected from theft and vandalism

Shadow Analysis

Using a solar path finder: The most accurate and convenient way to place the array away from any tall object that can possibly cause shadow is to use a solar pathfinder. When you use a solar pathfinder there is no need to measure distance or height of any objects that can createshadow at any point of time during the year.

Applying rule of thumb: The general rule of thumb is to locate the array at a distance away from the object that is at-least twice the height of the object. This will ensure that the object will not cast a shadow for four hours either side of solar noon.



Space between two rows: When PV modules are installed in multiple rows, consideration must be given so that one row of modules does not cast a shadow on the row behind. Calculations need to be done to find the minimum distance between PV array rows to avoid winter mid- day shading.

2. Mr. Y wants to design a PV system for a project, what are the design consideration he should take follow.

Design Considerations

- (1) Load bearing capacity of roof depends of the age of roof.
- (2) Typically, a RCC roof has a life of 100 years. Asbestos roof has much lower lifeat 30 years. Corrugated metal has life span between 20 and 100 years.
- (3) Usually the load of the structure including PV and the supporting structure varies from 30 kg/m² to 60kg/m².
- (4) As per MNRE technical specification the total load of the structure should be lessthan 60 kg/m².
- (5) For a large system a suitable walk-way will be required for maintenance purpose.
- (6) Operating temperature affects performance. Therefore, PV arrays should beinstalled in such way that their sufficient air flow/ventilation for cooling.
- (7) High wind pressure can damage the structure and modules. Therefore, mountingstructure should be opted such a way that there is minimum wind pressure.
- (8) High humidity and salty atmosphere can corrode the structure and the extremelevels in the site should be known.
- (9) Lightning strikes can damage the electrical equipment and sometimes the modules. Therefore, lightning vulnerability in the site should be evaluated.

3. Analyse the way in which mounting arrangements is to be done to get maximum electrical energy from the solar PV system.

Mounting the Solar Modules

- For mounting the solar panels first determine mounting method i.e. Roof mount or Ground mount.
- Modules should be oriented south facing to receive maximum sunlight.
- The Modules produce more power at low temperature and full sun.
- Tracking the Sun increases the amount of power from an array.
- Any obstruction (such as tree or building) should be avoided.
- The distance between solar panel and obstruction should be more than double the height of obstruction.

- The support for the Solar panel need to be a robust one and should not be accessible to general public.
- It should be so installed that rainwater, bird dropping, leaves etc. do not accumulate and the top surface can be cleaned easily.

4. Design solar pumping system for a 2 floor building of 15000 liters of water consumption every day. Daily water requirement 50 m³/ day from a depth of 20 m. It has elevation, standingwater level, and drawdown of 10 m, 10 m, and 4 m respectively.

- ✓ **Water density is 1000 kg/m³**
- ✓ **Acceleration due to gravity (g) is 9.8 m/s²**
- ✓ **The peak power rating of the solar module is 310 WP**
- ✓ **as the modules do not operate at its rated peak power capacity so the operating factor is 0.75.**
- ✓ **Pump efficiency is around 40 %**
- ✓ **Mismatch factor is 0.85 as the modules do not operate at the maximum PowerPoint.**

Determine the daily water requirement in (m³/day). Daily water requirement = 50 m³/day.

1 Calculate the Total Dynamic Head (TDH) required for pumping the water.

Total vertical lift = Elevation + Standing Water Level + Drawdown

Total vertical lift = 10 m + 10 m + 4 m = 24 m

Frictional loss = 5 % of the total vertical lift = $24 \times 0.05 = 1.2$ m

Total Dynamic Head (TDH) = Total vertical lift + Frictional loss

(TDH) = 24 m + 1.2 m = 25.2 m

2. Calculate the total hydraulic energy required per day (Watt-hour/day) for pumping the water.

Hydraulic energy required = Mass $\times$ g $\times$ TDH

Hydraulic energy required = Density $\times$ Volume $\times$ g $\times$ TDH

Hydraulic energy required = $1000 \text{ kg/m}^3 \times 50 \text{ m}^3/\text{day} \times 9.8 \text{ m/s}^2 \times 25.2 \text{ m}$ (multiply by 1/3600 to convert seconds in hours)

Hydraulic energy required = 3430 Wh/day

3. Calculate the solar radiation available at the site.

Solar radiation available at the site (No. of hours of peak sunshine per day) = 5.5h/day (1000 W/m² equivalent)

Peak sun hours are most commonly used as they simplify the calculations. Do not get confused with the “Mean Sunshine Hours” and “Peak Sun Hours” which you would collect from the meteorological station. The “Mean sunshine hours” indicate the number of hours the sunshine’s were as the “Peak sun hours” is the number of hours the actual amount of energy received in KWh/m²/day.

4. Calculate the size and number of PV modules required, the motor rating, its efficiency, and losses.

Total wattage of PV panel = Total hydraulic energy / No. of hours of peak sunshine per day

Total wattage of PV panel = 3430 / 5.5 = 624 W

Total wattage of PV panel considering system losses = $\frac{\text{Total wattage of PV panel}}{\text{Pump efficiency} \times \text{Mismatch factor}}$

Total wattage of PV panel considering system losses = 624 / (0.40 × 0.85) = 1836W

Total wattage of PV panel considering the operating factor of the PV module

$$= \frac{\text{Total wattage of PV panel considering system losses}}{\text{Operating factor}}$$

$$= 1836 / 0.75 = 2448 \text{ W}$$

No. of PV panels required of 310

$$WP = \frac{\text{Total wattage of PV panel considering the operating factor of the PV module}}{\text{No. of PV panels required of 310 Wp}}$$

$$= 2448 / 310 = 7.8 = (8 \text{ round figure})$$

Power rating of the DC motor = Total wattage of PV panel considering operating factor of the PV module / 746 W (i.e. 1 hp)

Power rating of the DC motor = 2448 / 746 W (i.e. 1 hp)

Power rating of the DC motor = 3.28 hp motor = 3.5 hp

5. Mr Y is supposed to go for installation of solar power plant. Which are the instruments and tools he need to carry with him for successful completion of solar installation? For what purpose they are needed?

➤ **Instruments:**

1) Digital multimeter: A Digital multimeter is used to measure two or more electrical values such as a voltage, current and resistance.

2) AC/DC Clampmeter: It combines digital multi meter with Current Sensor.

3) Hydrometer: It is used to find the state of charge in a battery by determining specific gravity of the electrolyte during maintenance.

4) Compass: Used for navigation orientation.

➤ **Tools used for installation:**

Wire strippers, Wire crimpers, Pliers, Screwdrivers, Spanners, Socket wrench, Hammer, Tester, Drilling machines, Knife.

6. Students of xyz polytechnic are interested to know the different stages of solar plant installation. Guide them by explaining the different steps of installation.

The installation of Solar Power System involves the following major steps:

- Civil Foundation Job
- Mounting of Solar Modules on the Support Structure.
- Installation of Battery Bank.
- Interconnection of SPV panel in series & parallel configuration, Charge Control Unit and FJB
- Connection of Inverter and Load
- Earthing of Lightning Protection Unit.

7. **Design a PV water pumping system, which is required to draw 25000 litres of water every day from a depth of 10 m. The data required for calculations is as follows:**

Water density = 1000 kg/m³

Acceleration due to gravity = $g = 9.8 \text{ m/s}^2$

The peak power rating of the solar module is 75 Wp

As the modules do not operate at its rated peak power capacity so the operating factor is 0.75.

Total vertical lift = 12 m (5 m- elevation, 5 m- standing water level, 2 m - drawdown)

Pump efficiency = 30%

Mismatch factor = 0.85 as the modules do not operate at the maximum PowerPoint

Effective sunshine hours = 6

Calculations for PV Water Pumping System

Step 1. Daily water requirement = 25000=25 m³/day

Step 2: Total vertical lift = Elevation + Standing Water Level + Drawdown

Total vertical lift = 5+5+2=12 m

Frictional loss = 5 % of the total vertical lift = $12 \times 0.05 = 0.6 \text{ m}$

Total Dynamic Head (TDH) = Total vertical lift + Frictional loss

Total dynamic head (TDH) = 12 + 0.6 ==12.6 m

Step 3: Hydraulic energy required = Mass $\times g \times$ TDH

Hydraulic energy required = Density $\times$ Volume $\times g \times$ TDH

Hydraulic energy required = $1000 \text{ kg/m}^3 \times 25 \text{ m}^3/\text{day} \times 9.8 \text{ m/s}^2 \times 12.6 \text{ m} = 857.5$

Wh/day (multiply by 1/3600 to convert seconds in hours)

Hydraulic energy required= 857.5.35 Wh/day

Step 4: Calculate the solar radiation available at the site.

Solar radiation available at the site (No. of hours of peak sunshine per day) = 6 h/day
(1000 W/m² equivalent)

Step 5: Calculate the size and number of PV modules required, the motor rating, its efficiency, and losses.

Total wattage of PV panel= (Total hydraulic energy)/ (No. of hours of peak sunshine per day x Pump efficiency × Mismatch factor x Operating factor)

Total wattage of PV panel = $857.5 / (6.0 \times 0.3 \times 0.85 \times 0.75) = 747.27 \text{ W}$

No. of PV panels required of 75 Wp = $747.27 / 75 = 9.96 = (10 \text{ round figure})$

Power rating of the DC motor = $747.27 / 746 \text{ W}$ (i.e. 1 hp)

Power rating of the DC motor = 1 hp motor.

8. Design solar pumping system for a depth of 20 m and 15000 liters of water consumption every day. Daily water requirement 15 m³/ day from a depth of 20 m. It has elevation, standingwater level, and drawdown of 10 m, 10 m, and 4 m respectively.

- ✓ **Water density is 1000 kg/m³**
- ✓ **Acceleration due to gravity (g) is 9.8 m/s²**
- ✓ **The peak power rating of the solar module is 310 WP**
- ✓ **As the modules do not operate at its rated peak power capacity the operating factor is 0.75.**
- ✓ **Pump efficiency is around 40 %**
- ✓ **Mismatch factor is 0.85 as the modules do not operate at the maximum PowerPoint. Determine the daily water requirement in (m³/day).**

1 Calculate the Total Dynamic Head (TDH) required for pumping the water.

Total vertical lift = Elevation + Standing Water Level + Drawdown

Total vertical lift = 10 m + 10 m + 4 m = 24 m

Frictional loss = 5 % of the total vertical lift = $24 \times 0.05 = 1.2 \text{ m}$

Total Dynamic Head (TDH) = Total vertical lift + Frictional loss

Total Dynamic Head (TDH) = 24 m + 1.2 m = 25.2 m

2. Calculate the total hydraulic energy required per day (Watt-hour/day) for pumping the water.

Hydraulic energy required = Mass × g × TDH

Hydraulic energy required = Density $\times$ Volume $\times$ g $\times$ TDH

Hydraulic energy required = $1000 \text{ kg/m}^3 \times 15 \text{ m}^3/\text{day} \times 9.8 \text{ m/s}^2 \times 25.2 \text{ m}$ (multiply by 1/3600 to convert seconds in hours)

Hydraulic energy required = 1029 Wh/day

3. Calculate the solar radiation available at the site.

Solar radiation available at the site (No. of hours of peak sunshine per day)
= 5.5h/day (1000 W/m² equivalent)

Peak sun hours are most commonly used as they simplify the calculations. Do not get confused with the “Mean Sunshine Hours” and “Peak Sun Hours” which you would collect from the meteorological station. The “Mean sunshine hours” indicate the number of hours the sunshine’s were as the “Peak sun hours” is the number of hours the actual amount of energy received in KWh/m²/day.

4. Calculate the size and number of PV modules required, the motor rating, its efficiency, and losses.

Total wattage of PV panel = (Total hydraulic energy) / (No. of hours of peak sunshine per day)

Total wattage of PV panel = $1029 / 5.5 = 187 \text{ W}$

Total wattage of PV panel considering system losses

= Total wattage of PV panel / (Pump efficiency $\times$ Mismatch factor)

Total wattage of PV panel considering system losses = $187 / (0.40 \times 0.85) = 550 \text{ W}$

Total wattage of PV panel considering the operating factor of the PV module

= Total wattage of PV panel considering system losses / Operating factor

= $550 / 0.75 = 735 \text{ W}$

No. of PV panels required of 310 WP = Total wattage of PV panel considering the operating factor of the PV module / 310

No. of PV panels required of 310 Wp = $735 / 310 = 2.3 = \text{(3 round figure)}$

Power rating of the DC motor = Total wattage of PV panel considering operating factor of the PV module / 746 W(i.e. 1 hp)

Power rating of the DC motor = $735 / 746 \text{ W (i.e. 1 hp)}$

Power rating of the DC motor= 0.98 hp motor = 1 hp

- 9. Design a PV water pumping system for daily water requirement of 7000 litres from a depth of 42m m. It has elevation, standing water level, and drawdown of 10 m, 30 m, and 2 m respectively.**

Water density is 1000 kg/m³ , Acceleration due to gravity (g) is 9.8 m/s²

The peak power rating of the solar module is 310 WP

As the modules do not operate at its rated peak power capacity so the operating factor is 0.8. Pump efficiency is around 30 %, Mismatch factor is 0.85 as the modules do not operate at the maximum PowerPoint, Effective sunshine hours = 5.

Step 1: Daily water requirement = 7000 litres = $7 \text{ m}^3/\text{day}$

Step 2: Total vertical lift = Elevation + Standing Water Level + Drawdown

Total vertical lift = $10 \text{ m} + 30 \text{ m} + 2 \text{ m} = 42 \text{ m}$

Frictional loss = 5 % of the total vertical lift = $42 \times 0.05 = 2.1 \text{ m}$

Total Dynamic Head (TDH) = Total vertical lift + Frictional loss

Total Dynamic Head (TDH) = $42 \text{ m} + 2.1 \text{ m} = 44.1 \text{ m}$

Step 3: Hydraulic energy required = Mass $\times$ g $\times$ TDH

Hydraulic energy required = Density $\times$ Volume $\times$ g $\times$ TDH

Hydraulic energy required = $1000 \text{ kg/m}^3 \times 7 \text{ m}^3/\text{day} \times 9.8 \text{ m/s}^2 \times 44.1 \text{ m}$

(multiply by 1/3600 to convert seconds in hours)

Hydraulic energy required= 840.35 Wh/day

Step 4: Calculate the solar radiation available at the site.

Solar radiation available at the site (No. of hours of peak sunshine per day) = 5.0 h/day (1000 W/m^2 equivalent)

Step 5: Calculate the size and number of PV modules required, the motor rating, its efficiency, and losses.

Total wattage of PV panel = (Total hydraulic energy)/ (No. of hours of peak sunshine per day
x Pump efficiency x Mismatch factor x Operating factor)

Total wattage of PV panel = $840.35 / (5.0 \times 0.3 \times 0.85 \times 0.8) = 823.87 \text{ W}$

No. of PV panels required of 80 Wp = $823.87/80 = 10.3 = (11 \text{ round figure})$

Power rating of the DC motor

= $823.87 / 746 \text{ W (i.e. 1 hp)}$

Power rating of the DC motor = 1.1 hp motor.

10. Mr. X wants to plan a project in a particular area, what are the different tools he should choose for site assessment.

The following tools are required for site assessment.

- A solar pathfinder/sun eye
- A compass
- A measuring tape/digital distance meter
- A camera
- An angle measuring equipment
- A notebook
- A working partner

11. Mr. X wants to test a project before commissioning. What performance PV standards he should use? Describe the nature of the tests.

The performance PV standards, namely IEC 61215 and IEC 61646 are used for the design of a PV module. Various parameters tested are as follows.

Diagnostic: Visual inspection, Hot spot.

Electrical: Insulation resistance, Wet leakage current

Performance test: P_{max} at STC and P_{max} at low irradiance.

Thermal test: Bypass diode test, Hot spot.

Irradiance: Outdoor exposure, UV exposure, Light soaking.

Environmental test: Effect of Temperature, Humidity, Damp

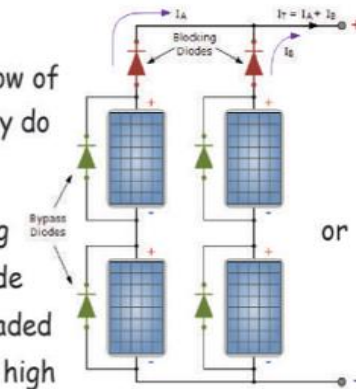
Mechanical: Mechanical load, Robustness of terminations, Hail impact

12. Discuss Blocking and Bypass diodes.

5.6.3 Blocking and bypass diodes

Blocking diode is essential to prevent the reverse flow of current at night from batteries to PV modules as they do not produce any voltage.

Bypass diodes: When one of the panels in the string few cells in the panel are shaded, bypass diodes provide an alternate path for the current to flow from unshaded area bypassing shaded panel or portion which has high resistance.



Feature	Blocking Diode	Bypass Diode
Connection Type	In series with the panel/string	In parallel with solar cell groups
Primary Function	Prevents reverse current flow	Bypasses shaded or damaged cells
Use Case	Nighttime protection	Shading and fault protection

13. Distinguish between wiring at AC component side and wiring at DC component side

Wiring at AC component side	Wiring at DC component side
AC voltage can easily be stepped up or stepped down using a Transformer.	In DC, voltages are added by connecting the solar modules in series. Connecting in parallel add up the current.
AC continuously changes polarity, which gives a repulsive shock and makes it easier to pull your hand away.	In DC, current continuously flows in one direction making it difficult to pull your hand away if you come in contact it.
There is no concept of a series connection since the polarity of AC output reverses constantly.	For a series connection, positive of source V1 is connected to negative of source V2.
Parallel connections in AC are made by using Synchronizers that match the polarity of one AC source to another AC source and then join them.	Parallel connections in DC are made by connecting positive terminals of all power sources together to get one positive output and all negative terminals are connected to give one negative output. The equivalent source voltage remains the same but the output current is a sum of current received from all power sources.

14. Design a standalone solar PV system consisting of inverter, batteries and PV panels for the loads, whose details are given in the table. Minimum sunshine hours may be assumed as 5 hrs. Missing data if any can be suitably assumed. Given:-Solar panels to be used are $W_p=80W$, $V_m=17.6V$ $I_m=4.55A$, system efficiency=0.8, Battery efficiency = 85%, Inverter efficiency =95%, Charge controller efficiency=95%, days of autonomy =1 DOD=70%

Load	Rating Watts	Quantity	Hours/day
CFL	20	5	4
Fan	80	5	4
TV	100	1	5
Computer	250	1	2
Refrigerator	150	1	10

Total load = $20 \times 5 + 80 \times 5 + 100 \times 1 + 250 \times 1 + 150 \times 1 = 1000 \text{ W}$

Total Energy = $20 \times 5 \times 4 + 80 \times 5 \times 4 + 100 \times 1 \times 5 + 250 \times 1 \times 2 + 150 \times 1 \times 10 = 4500 \text{ Wh}$

Sizing of inverter- Assuming 25 % extra load and 0.8 pf for inductive load

Inverter wattage = $(1000 \times 1.25)/0.8 = 1562 \text{ VA} \sim 1.5 \text{ KVA } 12V$

$$\text{Battery Sizing in Ah} = \frac{\text{Total energy in KWH} \times \text{No. of days of autonomy}}{\text{Battery efficiency} \times \text{DOD} \times \text{Nominal Battery Voltage}}$$

$$= (4500 \times 1) / (0.85 \times 0.7 \times 12)$$

$$= 630.25 \text{ Ah} \sim 600 \text{ Ah}$$

Let us choose 100 Ah 12V. Ah of the batteries remain at 100Ah when they are connected in series. To get 600Ah total 12 batteries are taken, 2 in series and 6 such sets in parallel.

Sizing of solar panel:

$$\text{Total wattage of panel} = \frac{\text{Total energy consumption /day}}{\text{Effective sunshine hours} \times \text{System efficiency}}$$

$$= 4500 / (5 \times 0.73) = 1232.87 \text{ w} \sim 1.25 \text{ KW}$$

We choose 80Wp 12V panels

We connect 2 panels in series to match 24V of inverter. 8 strings are connected in parallel.

15. Design an off-grid rooftop PV System for a two/three bed room residence building according to the load requirement.

Sl no	Item	Power in W	Numbers	Total Power in W	No.of hours operation	Wh/day
1	Tube lights	43	44	1892	6	11352
2	Ceiling Lamps	20	50	1000	6	6000
3	Ceiling Fans	140	35	4900	4	19600
Total				7792		36952

Total Energy Consumption/Day= 36.952 Kwh

Approximately Its **37 Units /Day Energy Consumption**

Total Energy Consumption/Month = 37X30 days= 1100 units

Requirement of PV system size to generate 37 Kwh

$$\text{Array rated power} = \frac{\text{Total energy consumption/day in Kwh}}{\text{Total sunshine hours X System efficiency}}$$

Location: Bangalore

Latitude:12.971° N

Longitude:77.59°E

solar radiation for Bangalore location is 5.5kWh/m²/day

$$\text{Array rated power} = \frac{\text{Total energy consumption/day in Kwh}}{\text{Total sunshine hours X System efficiency}}$$

$$\text{Array rated power} = \frac{37 \text{ Kwh}}{5.5 \text{ hr} \times 0.73} = 9.215 \text{ Kw} = 10 \text{ Kw}$$

Size of the PV system is 10Kw

- Module sizing and selection**

✓ Size of the PV system is 10 Kw

✓ System Voltage 24 V

Module Specifications

Electrical Data		CS6P-220M	CS6P-225M	CS6P-230M	CS6P-235M	CS6P-240M	CS6P-245M	CS6P-250M
Nominal Maximum Power at STC (Pmax)		220W	225W	230W	235W	240W	245W	250W
Optimum Operating Voltage (Vmp)		29.5V	29.7V	29.9V	30.1V	30.2V	30.3V	30.4V
Optimum Operating Current (Imp)		7.45A	7.58A	7.70A	7.82A	7.95A	8.09A	8.22A
Open Circuit Voltage (Voc)		36.9V	37.0V	37.1V	37.2V	37.3V	37.4V	37.5V
Short Circuit Current (Isc)		7.97A	8.07A	8.22A	8.34A	8.46A	8.61A	8.74A
Operating Temperature		-40°C~+85°C						
Maximum System Voltage		1,000V (IEC) /600V (UL)						
Maximum Series Fuse Rating		15A						
Power Tolerance		+5W						
Temperature Coefficient	Pmax	-0.45%/°C						
	Voc	-0.35 %/°C						
	Isc	0.060 %/°C						
	NOCT	45°C						

Under Standard Test Conditions (STC) of irradiance of 1000W/m², spectrum AM 1.5 and cell temperature of 25°C

For 10 kw system 250 Wp solar module selected

$$\text{Number of solar modules required} = \frac{\text{Array rated power}}{\text{Power of Selected solar module}}$$

$$\text{Number of solar modules required} = \frac{10000}{250} = 40$$

✓ Number of solar modules required 40 of 250 W.

Number of Strings in connected parallel are 10

In each string , Number of panels connected in series are 4

✓ 10 string X 4 panels = 40 panels

✓ 1 string-4 panels in series generates 1kw power

✓ 10 String=10 X 1 Kw= 10 Kw

● Battery sizing-

Size the batteries and system voltage for the estimated PV system according to the required backup

Battery Sizing Calculation:

1. Total Energy Demand
2. Battery Loss around 15%(Battery Efficiency)

3. Depth of discharge 40%
4. Nominal Battery Voltage
5. Days of Autonomy (No.of days backup)

$$\text{Battery Size} = \frac{\text{Total Energy Demand}}{\text{Battery Efficiency} \times \text{DOD} \times \text{Nominal Voltage}} \times \text{Days of Autonomy}$$

$$\text{Battery Size} = \frac{37000 \text{ wh}}{0.85 \times 0.6 \times 24 \text{ V}} \times 2$$

$$\text{Battery Size} = 6045.75 \text{ Ah}$$



Specification

Chemistry	LiFePO4
Nominal Voltage	25.6V
Nominal Capacity	200Ah
Nominal Energy	5120Wh
Max Continuous Charging Current	100A
Max Continuous Discharging Current	200A
Max. Load/ Inverter Power	4000W
Weight	88.1Lb/40Kg
Dimension(L*W*H)	20.5*10.5*8.6 inch 520*269*220mm
Wire(s) for Series & Parallel Connection / Loading	2AWG/35mm²

$$\checkmark \text{ Number of Battery Required} = \frac{6000}{200} = 30$$

- Size the charge controller for estimated battery system

Charge Controller is normally calculated by

- ✓ Battery bank Voltage
- ✓ Total solar panel power

$$\text{Charge Controller Current} = \frac{\text{Total solar panel power}}{\text{Battery bank Voltage}}$$

$$\text{Charge Controller Current} = \frac{10000}{24} = 417 \text{ A}$$

Sizing of inverters

Sl no	Item	Power in W	Numbers	Total Power in W	No.of hours operation	Wh/day
1	Tube lights	43	44	1892	6	11352
2	Ceiling Lamps	20	50	1000	6	6000
3	Ceiling Fans	140	35	4900	4	19600
Total				7792 watts		

Inverter sizing is based on total power of the load.

Size of the inverter = 25 % extra on total power of the load.

Size of the inverter for the system = Total power X 1.25

Size of the inverter for the system = 7792 X 1.25 = 9740

Size of the inverter required for the system is 10 KW or 10 KVA

16. Design 100 KW grid connected solar PV system without battery backup and estimate cost of installation, payback period and carbon footprint.

Grid connected 100 kw PV system without battery backup

1. Inverter selection -

Inverter sizing is based on total power of the load.

Size of the inverter = 25 % extra on total power of the load and pf =0.8 for inductive loads

Size of the inverter for the system = (Total power X 1.25)/0.8

= (100 x1.25)/0.8 = 156.25 KVA.

Therefore 160 KVA 240 V DC inverter is chosen.

2. Module sizing and selection

- Size of the PV system is 100 Kw
- System Voltage 24 V

Solar Panel chosen is 330W / 24V Poly

For 100 kw system 330 Wp solar module selected

Number of solar modules required = $100 \times 1000 / 330 = 304$ Panels ~ 310 panels

- Number of solar modules required 310 of 330 W.
- To match inverter voltage, no of panels in series = $10 \times 24 = 240$ V
- Number of Strings in connected parallel are $310/10 = 31$

Bill of materials for 100kW Solar PV System

Sl.No	Item	Specifications	Qty	Unit Cost Rs.	Total Cost
1	PV Module	330 Wp	310	13000	40,30,000
2	Solar Grid Inverter	160 KVA= 160 kw	1	12,00,000	12,00,000
3	Cables and Junction Boxes		1	Rs 3/ watt	3,00,000 (3 x 100x 1000)
4	Earthing and BoS + Accesories		1	Rs 7/watt	7 x 100 x 1000 = 7,00,000
Total					62,30,000

Approximate cost of the 100kw PV solar system is 62,30,000

Payback period

Average cost of state electricity supply in India, Indian rupees per kilowatt hour 6.15

On an average a solar system would generate 4 to 4.5 units per Kw, in India. So, a 100KW solar system would generate 400 kwh/day.

Yearly returns is $400 \times 365 \times 6.15 = 897900$ rs

Total initial investment of the 100Kw plant is **62,30, 000**

So, payback period will be $62,30,000 / 897900 = 7.5$ years

Calculate Carbon footprint

Total Coal Saved(1KG/unit) = 400 kg/day
Total Water Saved 3.3 Lts/Unit= $3.3 \times 400 = 1320$ Lts/day
Burning 1 kg of coal will produce about 3.3 kg of CO ₂ .
So $400 \text{ kg} \times 3.3 = 1320 \text{ kg/day}$ of carbon foot will be saved.

17. Mr Z is proceeding forward to install solar power plant. What are the safety measure do you recommend him to follow.

Safety measure:



Wear insulated safety gloves
Wear industrial shoes while working on the system



- Do not touch the panel junction box during day time, as DC voltage is always present when solar modules are exposed to sunlight.

- Off MCB in solar combiner box of the Solar Array before you work on them.



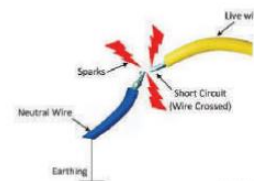
- Always check positive or negative polarity of any wire with Multimeter before connecting it to any device.



- Always connect positive wire in the last after completing all other connections, and remove positive wire first before starting any Maintenance Work, as Red Terminal is active hazardous wire.



- Do not leave any wire open; it can touch other terminals causing short-circuit.



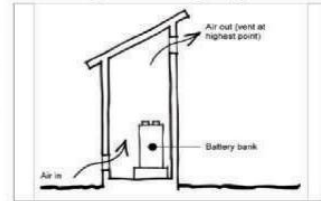
AC Side Wiring: In AC side wiring, a sequence of actions must be followed for the protection of components. AC output of Solar Hybrid Inverter must not be mixed with AC grid wire, as both of them are not in sync with each other, and can damage the system. We cannot connect neutral input and output of inverter to a common link. MCB on a AC wire must be less than the ampacity of that wire, which protects it from damage due to overload.

➤ Battery

Risk



Hydrogen gas is generated during charging of battery which can explode if a large amount of hydrogen gets accumulated in the battery room.



Safety Measure

- ✱ Provide a ventilation in the battery room for the gasses generated to move out.
- ✱ Place No Smoking warning sign outside.



Risk

Sulfuric acid present in batteries is harmful for the skin if it comes in direct contact. It can also cause eye burns.



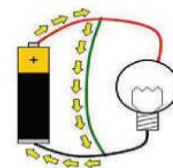
Safety Measure

- ✱ Wear safety glasses for protecting the eyes from the acid.
- ✱ If acid comes in contact with eyes, wash for 15 minutes and report to a doctor immediately.
- ✱ If acid comes in contact with skin or clothing, rinse off for several minutes and try not to spread the electrolyte. Report to a medic/doctor after rinsing.



Risk - Short Circuit

If two terminals of a battery are accidentally shorted, it can result in sparks and fire, and can also explode the battery.



Safety Measure

- ✱ Use connectors like MC4, insulated inline crimps and wire lugs for safe connections.
- ✱ While connecting batteries, take care that the two terminals



➤ Personal Safety



Risk

If you are working at a height, there is a risk of you falling down.

Safety Measure

Use safety harness, which will save you from falling

Risk

In a construction site, head injuries are common and cause fatal injuries.



Safety Measure

Use safety helmet while working with heavy loads like frame, battery, panels, etc.



Risk

The sulphuric acid in a lead acid battery is highly corrosive. Contact with eye can cause eye burns.



Safety Measure

Wear safety glasses for protecting the eyes from the acid.

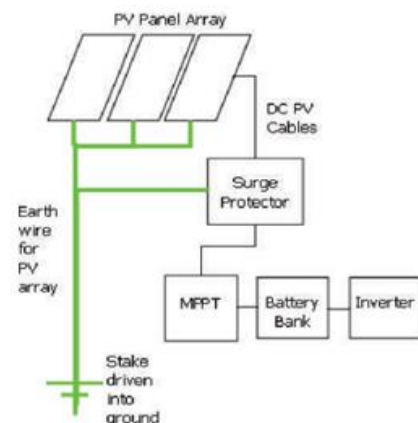
18. Discuss the importance of earthing of Solar PV System



The earthing is of great importance because it protects both the equipment as well as the persons using them. As the name implies the earthing means that the equipment or installations connected to the earth. The earthing is also called grounding.

The resistance of the earth is very low.

So when the equipment or installation develops some fault then the current will take the path through the earth and the equipment will remain safe. Therefore all power supply systems, consumer installations and electrical appliances are grounded as per Indian electricity rules 1972.



DC Voltage above 48 volts is dangerous, and should never be directly touched. As more number of DC power sources are connected in series, voltage of the system keeps on increasing. If string voltage of panels is above 48 volts DC, earthing must be provided to all the components for personal safety. Connect earthing wire of the respective building to one of the panels, and then interconnect all panels with green earthing wires of minimum thickness 0.75 mm.

19. Design a standalone solar PV system consisting of inverter, batteries and PV panels for the loads, whose details are given in the table. Minimum sunshine hours may be assumed as 5 hrs. Missing data if any can be suitably assumed. Given:-Solar panels to be used are $W_p=80W$, $V_m=17.6V$ $I_m=4.55A$, system efficiency=0.8, Battery efficiency = 85%, Inverter efficiency =95%, Charge controller efficiency=95%, days of autonomy =1 DOD=70%

Load	Rating in watts	Quantity	Hours/day
CFL	15	5	4
Fan	75	4	3
TV	100	1	3
Computer	250	1	2
Refrigerator	150	1	12

Total load = $15 \times 5 + 75 \times 4 + 100 \times 1 + 250 \times 1 + 150 \times 1 = 875 \text{ W}$

Total Energy = $15 \times 5 \times 4 + 75 \times 4 \times 3 + 100 \times 1 \times 3 + 250 \times 1 \times 2 + 150 \times 1 \times 12 = 3800 \text{ Wh}$

Sizing of inverter- Assuming 25 % extra load and 0.8 pf for inductive load

Inverter wattage = $(875 \times 1.25)/0.8 = 1367.18 \text{ VA} \sim 1.5 \text{ KVA } 12V$

Battery Sizing in Ah = (Total energy in KWH x No. of days of autonomy)

$$\begin{aligned} & \text{-----} \\ & \text{Battery efficiency} \times \text{DOD} \times \text{Nominal Battery Voltage} \\ & = (3800 \times 1) / (0.85 \times 0.7 \times 12) \\ & = 266 \text{ Ah} \sim 600 \text{ Ah} \end{aligned}$$

Let us choose 100 Ah 12V. Ah of the batteries remain at 100Ah when they are connected in series. To get 600Ah total 12 batteries are taken, 2 in series and 6 such sets in parallel.

Sizing of solar panel:

$$\text{Total wattage of panel} = \frac{\text{Total energy consumption /day}}{\text{Effective sunshine hours} \times \text{System efficiency}}$$

$$= 3800 / (5 \times 0.73) = 1041 \text{ w} \sim 1 \text{ KW}$$

We choose 80Wp 12V panels

We connect 2 panels in series to match 24V of inverter. 6 strings are connected in parallel.

Unit 3

1. **Suggest the safety precautions to be taken for their safety during inspection and maintenance of batteries.**

Safety precautions to follow during battery inspection and maintenance:

1. **Wear Protective Gear**

Always wear **safety goggles, gloves, and long-sleeved clothing** to protect against acid spills and electrical hazards.

2. **Disconnect Power Before Maintenance**

Always turn off the UPS or battery system before performing any inspection or maintenance to prevent electrical shocks.

3. **Work in a Well-Ventilated Area**

Ensure the area is well-ventilated to disperse any flammable gases (like hydrogen) released during battery operation or charging.

4. **Avoid Acid Contact**

If working with lead-acid batteries, avoid skin contact with sulfuric acid. Wear gloves and clean up any spills immediately using a neutralizing agent.

5. **Keep Tools Dry and Insulated**

Use **insulated** tools and ensure they are dry to avoid electrical shorts and reduce the risk of electrical shock.

6. **Avoid Overcharging**

Prevent overcharging by following manufacturer guidelines for charging levels and using proper charging equipment.

7. **Inspect for Damaged Batteries**

Check for signs of **leaks, bulges, or corrosion**. If the battery is damaged, handle it carefully and follow proper disposal methods.

8. **Have Emergency Equipment Nearby**

Keep a **fire extinguisher** (Class C for electrical fires) and a **first aid kit** on hand to respond to any accidents or emergencies quickly.

2. Discuss the main objectives of using SCADA

Objectives /Functions of SCADA:

1. **Monitoring:** Continuous monitoring of the parameters of voltage, current, etc.
2. **Measurement:** Measurement of variables for processing.
3. **Data Acquisition:** Frequent acquisition of data from RTUs and Data Loggers / Phasor data Concentrators (PDC).
4. **Data Communication:** Transmission and receiving of large amounts of data from field to control centres.
5. **Control:** Online real time control for closed loop and open loop processes.
6. **Automation:** Automatic tasks of switching of transmission lines, CBs, etc.

3. Differentiate between PLC and RTU

Feature	PLC (Programmable Logic Controller)	RTU (Remote Terminal Unit)
Primary Function	Local, real-time control of machinery and processes	Remote monitoring and control of equipment over long distances
Installation Environment	Typically used in industrial environments (factories, plants)	Used in remote or outdoor locations (substations, pipelines)
Programming	Programmed using ladder logic, function blocks, etc.	May have limited or no user programming; configured for communication
Connectivity	Communicates with local I/O devices	Communicates with central SCADA systems via telemetry
Power Requirements	Requires stable, continuous power supply	Designed for low power , sometimes powered by batteries or solar
Data Communication	May use Ethernet, Profibus, or Modbus for local networks	Uses protocols like DNP3, Modbus RTU, IEC 60870 for remote comms
Data Logging	Limited or no data logging	Often includes data logging and time stamping features
Maintenance	Requires on-site access for maintenance	Can be monitored and diagnosed remotely
Cost	Generally lower cost	Higher cost due to rugged design and communication features

4. **Discuss the necessity of renewable sources of electricity to charge Electric vehicle.**

The necessity of using **renewable sources of electricity** to charge **electric vehicles (EVs)**:

1. **Reduces Carbon Emissions**

Charging EVs with renewable energy (solar, wind, hydro) minimizes greenhouse gas emissions, making transportation truly eco-friendly.

2. **Maximizes EV Environmental Benefits**

EVs produce no tailpipe emissions, but using fossil fuels for charging offsets this advantage. Renewables ensure clean energy use throughout.

3. **Improves Air Quality**

Renewable energy doesn't emit harmful pollutants, unlike coal or gas power plants, leading to cleaner air and better public health.

4. **Supports Energy Independence**

Renewable sources reduce reliance on imported fossil fuels, enhancing national energy security and economic stability.

5. **Enables Sustainable Grid Integration**

EVs powered by renewables support the development of smart, decentralized, and sustainable energy grids.

6. **Promotes Long-Term Cost Savings**

Though initial investment is high, renewable energy offers low operating costs and price stability over time for EV charging.

5. **Analyse the necessity of Power Converters for charging Electric Cars from PV and Wind.**

Charging electric vehicles directly from **solar photovoltaic (PV)** and **wind energy systems** requires **power converters** to ensure compatibility, efficiency, and safety. These converters are essential for managing variable output from renewable sources and converting it into a stable form suitable for EV batteries.

Power Converters Are Necessary:

1. **To avoid Mismatch in Voltage and Current**

- PV panels and wind turbines produce **DC or variable AC voltages**, which are not directly compatible with EV battery charging systems.

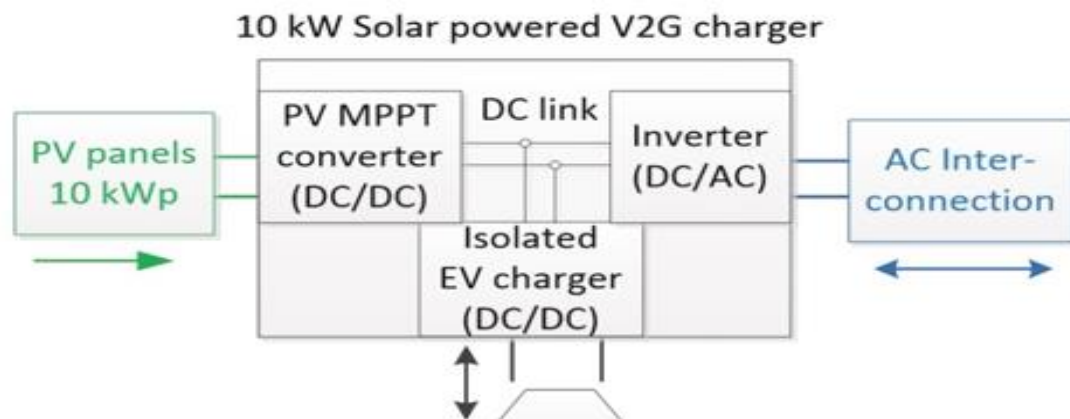
- Power converters **regulate and match** the voltage/current levels to meet battery specifications.

2. To avoid Intermittency of Renewable Sources

- Solar and wind outputs fluctuate with weather conditions.
- Converters with **Maximum Power Point Tracking (MPPT)** optimize power extraction and maintain efficiency.

3. Energy Storage Integration

- Power converters manage the flow between **renewables, batteries (if used), and EVs**, enabling energy storage when demand is low and discharge during charging.



From the point of view of power conversion, the generators used in wind turbines usually produce variable frequency AC electricity. Two back-to-back AC-to-DC and DC-to-AC power converters used to convert variable frequency AC power to high voltage or medium voltage 50 Hz or 60 Hz AC power for long-distance power transmission are done. This power is then moved to low voltage AC power, and the EV can then be charged using either AC or DC charging.

The simplest way to realize a solar-powered EV charging station is to use a solar inverter. The DC-to-DC power converter operates the solar panels at maximum PowerPoint. Then, the DC-AC inverter converts the DC power into 50 Hz or 60 Hz AC power for AC charging of the EV. However, there is a disadvantage to this method. Both photovoltaic panels and EV batteries are direct current or DC in nature. And in this method, DC power is inexplicably converted to AC and back. Therefore, a more efficient way to charge an EV from a PV is to use an isolated DC-to-DC converter and directly charge an EV from a PV using DC charging as shown in the

figure

6. Discuss the different modes of EV charging as per the standard IEC 61851

The IEC 61851 standard has defined different modes for EV charging:

- **Charging mode 1** uses regular 230 V, AC sockets. Mode 1 is limited to a maximum charging capacity of 2.3 kW to ensure safety as it lacks communication.
- **Charging mode 2** uses an In-Cable Control Box (ICCB) with a regular 230 V socket. The ICCB controls the charging capacity. Although normally used at 2.3 kW (1-phase, 10 A), Mode 2 has a maximum charging capacity of 7.4 kW (single phase, 32 A) or 22 kW (three-phase, 32 A)
- **Charging mode 3** determines the adequate charging capacity by communication between EV and charging point. Most mode 3 public charging points deliver 11 kW, 22kW or fast charging.
- **Charging mode 4** is DC charging and is usually applied for fast charging. Charging capacity ranges from 50 kW and higher.

7. Analyse and suggest the steps to be followed to maintain the batteries

Use the following as a guide when examining your battery:

1. Check the battery's state of charge. Most batteries have a State of Charge Indicator on top of the battery that will give you an on the spot diagnosis of the battery condition. However, a more reliable way to check is with a voltmeter to determine the stabilized voltage or if the vent caps are removable a hydrometer to determine the specific gravity (SG) of the electrolyte. A charged Century battery will have a stabilized voltage above 12.5 volts and an SG reading above 1.240.
2. Ensure the battery top is clean, dry, free of dirt and grime. A dirty battery can discharge across the grime on top of the battery casing.
3. Inspect the terminals, screws, clamps, and cables for breakage, damage or loose connections. These should be clean, tight and free of corrosion.
4. Apply a thin coating of high-temperature grease to posts and cable connections for added protection.
5. Inspect the battery case for obvious signs of physical damage or warpage. This usually indicates the battery has been overheated or has been overcharged.

6. If you have a maintainable battery, it is important to check if the battery has sufficient electrolyte covering the battery plates. If topping up is required, do not overfill as the fluid levels will rise when the battery is fully charged and may overflow. Top up using distilled or demineralized water and never fill with sulphuric acid.

7. When servicing a sealed maintenance free (SMF) battery, check the State of Charge Indicator. This gives you a snapshot of the battery's condition and whether the battery needs to be charged or replaced. The vehicle may still start the engine although the indicator outlines to replace the battery. If the State of Charge Indicator advises 'Replace Battery' it is important that the battery is replaced as the electrolyte levels may be below the plates which can lead to an internal explosion.

8. For batteries used in seasonal applications and stored long term, fully recharge the battery prior to storing. Check the state of charge or voltage regularly. Should the voltage drop below 12.5V, recharge the battery. It is important to check the battery completely before reconnecting to electrical devices.

If you are unsure about the condition or state of charge of the battery, take it to your local Century battery reseller. They can inspect and test your battery and provide you with professional advice and assistance.

8. For solar plants capacity 2GW, how will you optimize operation and maintenance process. Suggest any innovative method to increase cleaning cycle 24 to 365, to increase operational efficiency.

The traditional method of cleaning modules uses water to wash them manually using human labour.

- Water scarcity, expensive labour, and solar projects located in arid, drought-prone regions have made this process expensive and unviable. To tackle this issue, the solar industry has been shifting to robotic cleaning technologies, which provide a cost-effective method to clean solar panels compared to manual cleaning.
- Robotic cleaning solutions are gaining acceptance amongst utility-scale solar developers because of their efficiency and cost-effectiveness.
- Robotic Cleaning Can Maximize Energy Yield & Reduce O&M Cost of Solar Projects.

9. Discuss the steps to install solar at-home EV charger

Step 1: Choose an Installation Spot

The best location for solar panels is on a south-facing spot on your roof that will get the most sunlight. If you are only installing the panels for your EV charger, your garage can be a good place to install it, free of shadows and overhanging trees.

Step 2: Mount the Brackets

Solar panels need to be mounted securely to your roof or garage using sturdy brackets. Since solar panels are fragile and expensive, it's important to make sure they're mounted correctly. Solar panel brackets usually come with the solar panel kit, but if not, you can purchase them at a hardware store.

Step 3: Install the Panels

The panels should now be carefully placed and angled appropriately at around 18 to 36 degrees depending on the slope of the roof. These will be secured to the brackets using bolts tightened enough to be strong but not too tight as to strip the bolts.

Step 4: Get the Electrical Wired

An electrician can now do the electrical wiring that will allow you to harness the solar power in your appliances. This will include connecting the solar panel to an inverter and then to your home's electrical grid.

Step 5: Connect the Solar Inverter and Solar Battery

Your solar panels require an inverter and a battery to convert and harness the energy it gathers from UV rays. The solar inverter changes the direct current (DC) electricity that solar panels create into an alternating current (AC).

This is the type of energy used by most appliances and devices in your home. The solar battery stores solar power so that it can be used when there is no sunlight, like at night or on cloudy days.

Step 6: Install the Electric Car Charger Per Manufacturer Instructions

And lastly, install your new electric vehicle charger according to its instructions. Your newly installed solar-powered EV charger will use the solar panels to generate electricity, which is then used to charge your electric vehicle.

10. For 2 GW solar plant which communication protocol you recommend for SCADA.

Justify your recommendation. Can you Suggest typical SCADA hardware and software required for the plant?

Most widely protocol for communication used in SCADA is MODBUS

- MODBUS allows communication between several devices connected to the network through RTUs in SCADA.
- In this protocol, the master may initiate a MODBUS command to activate the connected element / device. The command contains the MODBUS address of the device. Only the indented device will act on receiving the command though other devices receive them. The MODBUS command ensures that the command is received unchanged through an inbuilt checking information.
- Easy installation, configuration and use
- Open specifications that do not require hardware constraints.
- Reliable communication between automation devices
- Interoperability between devices from different manufacturers.
- The use of the Modbus RTU protocol is compatible with many other industrial automation products: PLCs, temperature controllers, operator panels, data loggers, etc., which can easily communicate with a common supervisor.

Hardware

SCADA mainly consists of RTUs, Central processor and associated memory, Communications and Other equipment, etc.,

Remote Terminal Units (RTU)

The RTU provides an interface to the field analog and digital sensors situated at each remote site.

RTU Hardware:

The RTU Hardware consists of inputs, outputs, memory and communications.

Inputs

- Analog inputs (AI)
- Counter inputs (CI)
- Digital inputs (DI)

Outputs; Analog outputs (AO), Digital outputs (DO)

Memory: Static and Dynamic Memory

Control processor and associated memory, Communications and Other equipment, Communication interface(s) Power suppl, RTU rack and enclosure.

Software functions

- Real time operating system.
- Driver for the communications system, which is the link to the SCADA Master.
- Device drivers for the I/O system connecting to the field devices.
- SCADA applications like scanning of inputs, processing and storing of data, responding to requests from the SCADA master over the communications network.
- Some method may exist to allow the user applications to be configured in the RTU. This may be simple parameter setting, enabling or disabling specific I/O's or it may represent a complete user programming environment.
- Some RTU's may have a file system with support for file downloads. This supports user programs, and configuration files.

11. Smart cities are planned for city X, Discuss the different areas in which SCADA can be used so that better utilisation of resources

Electric power system, operation and control: SCADA systems are used in electric power generation plants, transmission area and distribution system.

Manufacturing Industries or plants: A SCADA helps in management of different inventory items or raw materials, controlling of automated systems in synchronous manner.

Telecom and IT based systems: - Management of different RF based systems, communication mediums and large communication systems including data logging through antennas can be easily done through the SCADA.

Water and sewage treatment plants and supply management: - SCADA based systems are used by the state or municipal corporation to monitor, control and regulate water capacities in reservoir.

Traffic controls: SCADA helps in regulation of traffic signals, controls the traffic flow in railway systems on road systems and air traffic controls.

Lift and Elevator controls: SCADA can also be used for lift and elevator controls.

12. Tesla model 3 electric car has 50kWh battery capacity and 350 kms range. Estimate the amount of Electricity needed to charge this vehicle? Assuming that the owner of the vehicle travels 2000 kms /month, calculate the number of solar panels needed. Missing data can be suitably assumed.

Distance travelled per month = 2000 kms.

Tesla 3 can drive 350 kms on a single charge.

Number of full charges needed per month can be calculated as,

$$2000 \text{ kms} / 350 \text{ kms per charge} = 5.71 \text{ approx } 6 \text{ charges per month}$$

Therefore, energy needed per month is,

$$50 \text{ kWh battery} \times 6 \text{ charges per month} = 300 \text{ kWh needed per month}$$

Energy needed per day:

$$300 \text{ kWh per month} / 30 \text{ days in a month} = 10 \text{ kWh needed per day}$$

Total wattage of solar panels = Total energy required per day/ (effective Sunshine hours x panel efficiency)

$$= 10 \text{ kWh per day} / (5 .5 \times 0.78) = 2.33 \text{ kW}$$

Choosing 400W or 0.4 kW solar panel, the number of solar panels needed will be,

$$2.33 \text{ kW} / .4 \text{ kW} = 5.82 \text{ approx } 6 \text{ solar panels}$$

Rounding up, 6 solar panels of 400W each are needed to charge the Tesla 3 with the long-range battery.

- 13. Tesla model 3 electric car has 70kWh battery capacity and 500 kms range. Estimate the amount of Electricity needed to charge this vehicle? Assuming that the owner of the vehicle travels 2000 kms /month, calculate the number of solar panels needed. Missing data can be suitably assumed.**

Distance travelled per month = 2000 kms.

Tesla 3 can drive 500 kms on a single charge.

Number of full charges needed per month can be calculated as,

$$2000 \text{ kms} / 500 \text{ kms per charge} = 4 \text{ charges per month}$$

Therefore, energy needed per month is,

$$70 \text{ kWh battery} \times 4 \text{ charges per month} = 280 \text{ kWh needed per month}$$

Energy needed per day:

$$280 \text{ kWh per month} / 30 \text{ days in a month} = 9.3 \text{ kWh needed per day}$$

Total wattage of solar panels = Total energy required per day/ (effective Sunshine
hours x panel efficiency)

$$= 9.3 \text{ kWh per day} / (5.5 \times 0.78) = 2.38 \text{ kW}$$

Choosing 400W or 0.4 kW solar panel, the number of solar panels needed will be,

$$2.38 \text{ kW} / 0.4 \text{ kW} = 5.95 \text{ solar panels}$$

Rounding up, 6 solar panels of 400W each are needed to charge the Tesla 3 with the long-range battery.

- 14. A company has got O & M work for a PV project, what are the different standard operating procedure they will have to follow.**

Processes involved in the maintenance of solar power systems

- The timely and regular cleaning of solar cells and PV panels
- Regular maintenance of all thermal-based components
- Servicing of HT side equipment on an annual basis
- Testing and upkeep of circuits
- Tracing of IV curves and thermal imaging
- Measure of earth resistance value
- Retro-commissioning (fine-tuning)
- Management of warranties
- System checks pertaining to data acquisition, etc.

- 15. Discuss the precautions to be taken to increase the life span of a Battery.**

- Do not allow the battery to stand idle for long time, this may cause inactivation of the battery cells.
- Do not charge the battery with very high rate of current because high rate of charging causes high temperature rise and excessive gassing, resulting in heavy loss of water and sometime overflowing of electrolyte from the battery cells.
- After every complete discharge, the battery should be immediately charged before returning it to its regular floating service. Otherwise there may be a chance of deposition of sulphate film on the plates.
- As already mentioned, the battery cells should be charged gently at normal rate so that there will be no chance of immediate gassing and temperature rise over 40°C. Otherwise there may be damage to the battery cells due to high temperature.
- During charging of batteries, continuous monitoring is required, if gassing starts and the temperature reaches the said limit, then reduce the rate of charging. If after reducing the charging rate, the temperature is still approaching to the limit, it indicates the completion of charging process because even the normal rate of charging may produce high temperature rise if the battery approaches to the fully charged condition.

- The voltage of each of the battery cells should be checked before conclusion of charging and each of the cells are properly and equally charged and the readings should also be tallied with previous record.
- The battery should be refilled with fresh electrolyte once in three years interval to maintain the battery in proper operating conditions.

16. Discuss how EVs can be utilised for better RE Grid Integration

EVs can be used to improve RE grid integration in several ways.

- It can stabilize the grid by smoothing out fluctuation in electricity demand.
- Vehicle to grid charging can reduce the cost of reinforcing local electricity grids by lowering demand peaks.
- By powering EV'S with RE, we can reduce Greenhouse gas emissions.
- By using RE for EV'S, the cost can be reduced and it is environmentally friendly.
- Integrating RE with EV charging stations can help in building positive public attention and can promote renewable energy adoption.
- EV vehicles can be used as micro grid to provide power to buildings and neighbourhoods during power outages or during emergency conditions.

Unit 4

1. Differentiate between On Grid Wind Turbine and Off Grid Wind Turbine

Feature	On-Grid Wind Turbine	Off-Grid Wind Turbine
Connection	Connected to the public electricity grid	Not connected to the electricity grid
Power Usage	Supplies electricity to the grid or home/building directly	Supplies power only to the local system or storage
Power Storage	Usually does not require batteries	Requires batteries or another form of energy storage
Excess Energy	Sent to the grid (can earn credits via net metering)	Stored in batteries or wasted if storage is full
Energy Backup	Grid acts as backup when wind is insufficient	Must rely on stored energy or a backup generator
Installation Cost	Generally lower (no need for storage systems)	Higher due to battery and inverter costs
Use Case	Urban areas, homes/buildings with grid access	Remote areas, rural applications without grid access
Reliability	Highly reliable due to grid backup	Less reliable unless battery backup is robust

Off-Grid Wind Turbine

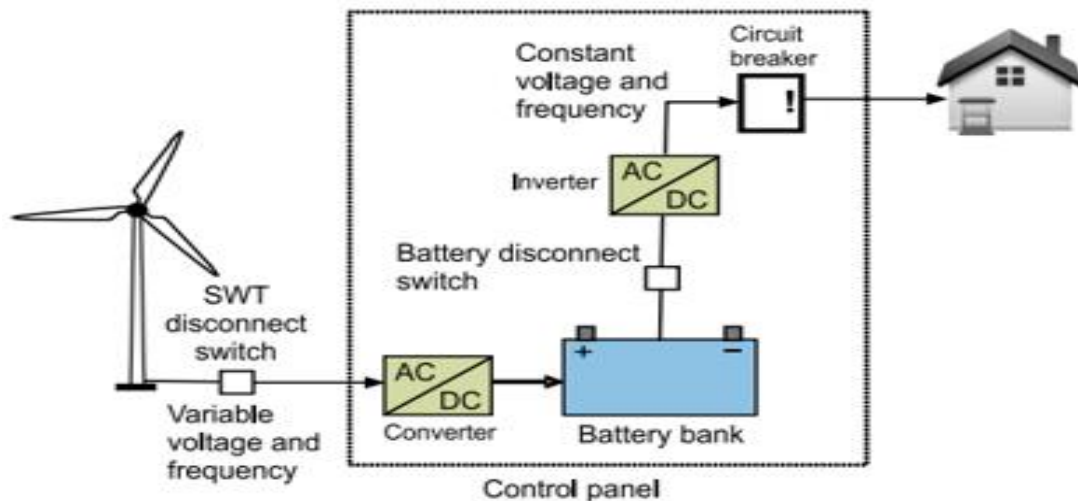


Fig 1

Off-grid refers to a SWT operating without connection to the public electrical grid network. Stand-alone SWTs which operate without being connected to the national grid are also called isolated grid systems. Stand alone power systems (SAPS) are off-grid electricity generation systems which may include wind power, solar photovoltaics, microhydro turbines,

battery bank and sometimes, a backup diesel generator. They are also commonly used at mountain tops, islands, single residential (see Figure 1), multi-residential, on boats or remote stations. SWTs can help in reducing the use of diesel generators, thereby saving fuel costs and reduce pollution.

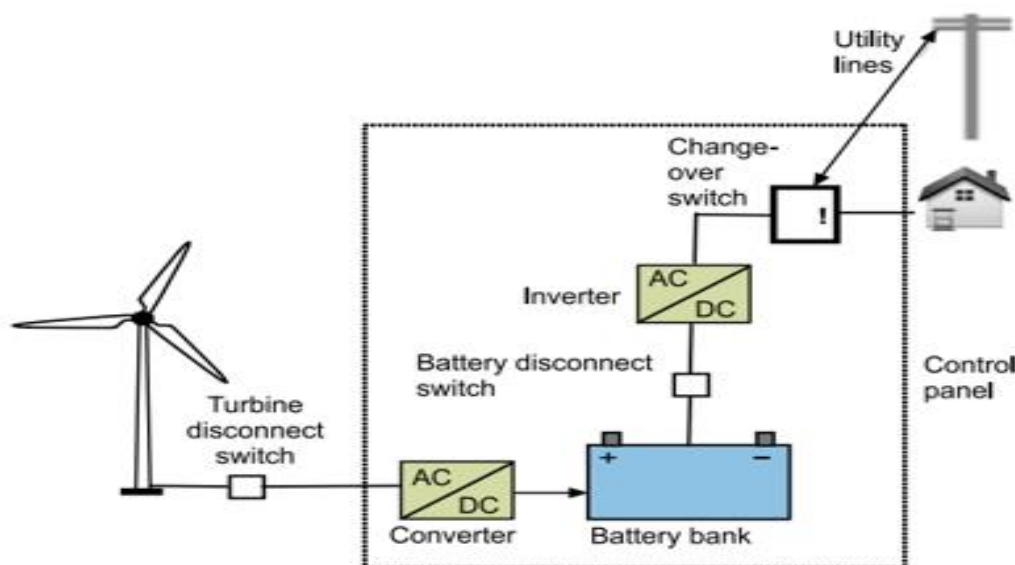
Since wind is usually a varying energy resource, off-grid systems are typically installed with some form of energy storage device (usually, a battery bank) that stores excess wind-generated electricity and supplies it to the load (e.g. homes) when there is insufficient wind.

If the SWT has a DC generator, it is used directly for battery charging. Alternatively, an inverter can convert DC power into AC power of the same quality as grid power. A battery bank can supply reserve power during calm spells. The advantage of such systems is that it is easy to set up. But the disadvantage is that the batteries are expensive and require regular maintenance.

On Grid Wind Turbine:

An **On-Grid Wind Turbine** is a wind energy system connected to the public electricity grid. It generates power from wind and supplies it directly to the grid or to a home/building through an inverter. Key points include:

- **Connection to Grid:** Directly tied to the local utility grid.
- **No Need for Batteries:** Power can be used instantly or fed into the grid; excess energy earns credits (net metering).
- **Cost-Efficient:** Lower initial cost as battery storage is not needed.
- **Backup from Grid:** If wind generation is insufficient, power can be drawn from the grid.
- **Use Case:** Ideal for areas with reliable grid access, like cities or towns.



2. A WPP of 80 m rotor diameter is rotating at a particular windy site with the average wind speed of 6 m/s at a power co-efficient of 0.4. Assuming the air temperature as 25°C with a density of 1.225 kg/m³, calculate the (a) Total power available in the wind (b) maximum power density (c) actual power density (d) power output from WPP

$$\begin{aligned}\text{(a) Power density of air } P_{\text{total}} &= \frac{1}{2} \rho v^3 \\ &= \frac{1}{2} \times 1.225 \times 6^3 \\ &= 132.3 \text{ W/m}^2\end{aligned}$$

(b) Maximum possible power density is limited by the Betz limit of 0.593.

$$\begin{aligned}\therefore P_{\text{max}} &= 0.593 \times P_{\text{total}} \\ &= 0.593 \times 132.3 = 78.45 \text{ W/m}^2\end{aligned}$$

(c) Actual power density is decided by the efficiency of the blades, i.e., 40%.

$$\begin{aligned}\therefore P_{\text{actual}} &= 78.45 \times 0.4 \\ &= 31.38 \text{ W/m}^2\end{aligned}$$

(d) Power output from the WPP = $P_{\text{actual}} \times \text{Swept area}$

$$\begin{aligned}&= 31.38 \times \frac{\pi D^2}{4} \\ &= 31.38 \times 4024 \text{ W} \\ &= 157653/1000 \text{ kW} \\ &= 157.653 \text{ kW}\end{aligned}$$

3. Government would like to make huge invest on small wind turbine across the state. As a renewable pathway student let them know the application this creates. (OR)

List the applications of small wind turbine?

1. Signalling
2. Sailboats
3. Street lamps
4. Remote houses
5. Farms
6. Water pumping
7. Seawater desalination
8. Village power
9. Mini grid
10. Industrial
11. Public centers
12. Car parkings

4. Mr B would like to be an entrepreneur to start a biomass plant. Suggest the different types of feedstock which he can collect.

Biomass is organic matter that comes from living organisms.

1. Agricultural Residues

- Rice husks
- Wheat straw
- Corn stalks and cobs
- Sugarcane bagasse
- Coconut shells
- Cotton stalks

2. Forestry Residues

- Wood chips
- Sawdust
- Tree bark
- Logging residues (branches, stumps)

3. Energy Crops

- Switchgrass
- Miscanthus
- Willow
- Jatropha
- Napier grass

4. Animal Waste

- Poultry litter
- Cow dung
- Pig manure

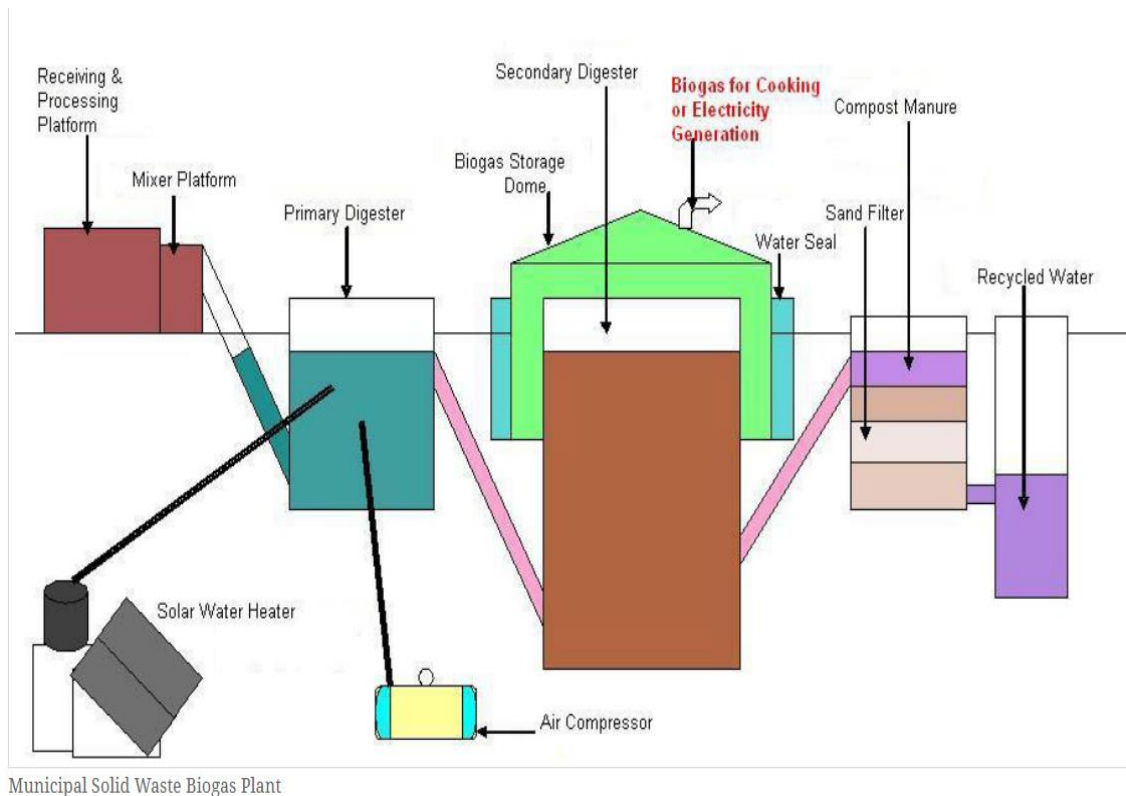
5. Industrial Organic Waste

- Food processing waste
- Paper mill sludge
- Brewer's grain
- Molasses

6. Municipal Solid Waste (Biodegradable)

- Kitchen waste
- Garden/yard waste
- Sewage sludge

5. Discuss Municipal waste-based Biogas Plant operation with block diagram



The segregated wet garbage (food waste) is brought to the plant site in bins and containers. It is loaded on a sorting platform and residual plastic, metal; glass and other non-biodegradable items are further segregated. The waste is loaded into a Waste Crusher along with water, which is mounted on the platform. The food waste slurry mixed with hot water is directly charged into the Primary digester.

This digester serves mainly as hydrolysis cum acidification tank for the treatment of suspended solids. For breaking slag compressed air is used for agitation of slurry. Compressed air will also help in increasing aeration since bacteria involved in this tank are aerobic in nature. The tank is designed in such a way that after the system reaches equilibrium in initial 4-5 days, the fresh slurry entering the tank will displace equal amount of digested matter from top into the main digester tank.

Main digester tank serves as a methane fermentation tank and BOD reduction takes place here. The treated overflow from this digester is connected to the manure pits. This manure can be supplied to farmers at the rate of Rs.4-5 per Kg. Alternatively municipal gardens and local gardens can be assured of regular manure from this biogas plant.

The biogas is collected in a dome (Gas holder) is a drum like structure, fabricated either of mild steel sheets or fibreglass reinforced plastic (FRP). It fits like a cap on the mouth of digester where it is submerged in the water and rests in the ledge, constructed inside the digester for this purpose. The drum collects gas, which is produced from the slurry inside the digester as it gets decomposed and rises upward, being lighter than air. Biogas burners will be provided. The biogas can be used for cooking, heating and power generation purpose.

6. Analyse the functions of the following parts of wind turbine generator a. Gear Mechanism b. Anemometer c. Yaw mechanism d. Nacelle e. Sensors

a. Gear Mechanism (Gearbox)

- Converts the **low rotational speed** of the turbine rotor (typically 10–60 rpm) to a **higher speed** (1,000–1,800 rpm) suitable for **electricity generation** in the generator.
- It ensures the generator operates within its optimal speed range for efficient power output.

Note: Some modern turbines use **direct-drive systems** that eliminate the gearbox.

b. Anemometer

- Measures **wind speed**, and sometimes direction.
- Provides real-time wind data to the turbine's control system.
- Helps in:
 - Starting or stopping the turbine when wind speed is within or outside operational limits.

c. Yaw Mechanism

- Rotates the **nacelle** (the housing at the top of the tower) to keep the **rotor facing the wind direction**.
- The yaw system of a WPP consists of the yaw drive and the yaw bearing and is located at the bottom of the nacelle mainframe. Depending on the wind direction sensed by the wind vane and the wind speed sensed by the anemometer, the electronic controller commands the yaw drive to turn the nacelle to face the wind.

d. Nacelle

- Encloses and protects **key components** such as the **gearbox, generator, main shaft, brake system, yaw mechanism, and control systems**.
- Provides structural support and accessibility for maintenance.

- Acts as the **central housing unit** of the wind turbine's mechanical and electrical systems.
- It is made of GRP or thin steel.

e. Sensors

For a typical multi megawatt WPP, there could be around 500 sensors.

- Monitor various **parameters** critical for turbine performance, safety, and maintenance.
Common types of sensors include:
 - **Temperature sensors:** Monitor bearings, gearbox, generator.
 - **Vibration sensors:** Detect mechanical imbalances or wear.
 - **Position sensors:** Detect rotor and blade positions.
 - **Wind sensors** (e.g., wind speed/direction via anemometers and vanes).
 - **Strain gauges:** Monitor stress on blades or tower.
- Feed data into the **control system**, enabling real-time adjustments and condition monitoring (often used in predictive maintenance).

7. Calculate the power developed and speed in rpm for a small wind turbine generator with the following particulars. Rotor efficiency =40% , Generator efficiency =70 % , Rotor swept area =2.11 m², Wind speed = 8.6 m/s, Tip Speed Ratio =7, Radius of rotor = 0.82

Given data: **Rotor efficiency (C_p)** = 40% = 0.40 **Generator efficiency (η_g)** = 70% = 0.70

Rotor swept area (A) = 2.11 m² **Wind speed (V)** = 8.6 m/s **Tip Speed Ratio (TSR)** = 7

Rotor radius (R) = 0.82 m

Step 1: Calculate Wind Power Available

The total wind power available is given by the formula:

$$P_{\text{wind}} = \frac{1}{2} \cdot \rho \cdot A \cdot V^3$$

Assume **air density (ρ)** = 1.225 kg/m³ (standard value at sea level).

$$P_{\text{wind}} = \frac{1}{2} \cdot 1.225 \cdot 2.11 \cdot (8.6)^3$$

$$P_{\text{wind}} \approx 0.6125 \cdot 2.11 \cdot 636.056$$

$$P_{\text{wind}} \approx 1.2924 \cdot 636.056 \approx 822.61 \text{ watts}$$

Step 2: Calculate Power Extracted by Rotor

$$P_{\text{rotor}} = P_{\text{wind}} \cdot \text{Rotor efficiency}$$

$$P_{\text{rotor}} = 822.61 \cdot 0.40 \approx 329.04 \text{ watts}$$

Step 3: Calculate Power Delivered by Generator

$$P_{\text{gen}} = P_{\text{rotor}} \cdot \text{Generator efficiency}$$

$$P_{\text{gen}} = 329.04 \cdot 0.70 \approx 230.33 \text{ watts}$$

Step 4: Calculate Rotor Speed (in rpm)

Tip Speed Ratio (TSR) is defined as:

$$\text{TSR} = \frac{\omega \cdot R}{V}$$

Where:

- ω is angular speed in rad/s
- $R = 0.82 \text{ m}$
- $V = 8.6 \text{ m/s}$

$$\omega = \frac{\text{TSR} \cdot V}{R} = \frac{7 \cdot 8.6}{0.82} \approx \frac{60.2}{0.82} \approx 73.41 \text{ rad/s}$$

Convert rad/s to rpm:

$$\text{rpm} = \omega \cdot \frac{60}{2\pi} \approx 73.41 \cdot \frac{60}{6.283} \approx 73.41 \cdot 9.55 \approx 701.58 \text{ rpm}$$

Final Answers:

- **Power output (electrical): ≈ 230.33 watts**
- **Rotor speed: ≈ 702 rpm**

8. Discuss the different methods of producing bio ethanol

a) From first generation Biomass

1. The first generation bioethanols refer to the ethanols that have been derived from the food-based feedstocks, such as starch, sugar, etc
2. These fermentable sugars are extracted by grinding or crushing followed by fermentation to ethanol.

b) From non-feed stocks

1. The second generation bioethanols are those produced from non-food feedstocks.
2. The non-food based feedstocks used for production of second generation ethanol comprises of dedicated energy crops (e.g., switchgrass,) and agricultural and wood residues (e.g., woodchips, sugarcane bagasse, and sawdust).
3. Cellulosic biomass is first pretreated for microbial fermentation to produce ethanol.

c) From Algae biomass

1. The third-generation bioethanol is obtained from the algal biomass. Algae can be used to produce a variety of biofuels such as hydrogen, diesel, isobutene, and ethanol.
2. Microalgae are unicellular organisms that are either autotrophic (produce their own food) or heterotrophic and can grow in diverse environment.
3. Autotrophic algae use sunlight and atmospheric CO₂ to form carbohydrates such as **starch and cellulose via photosynthesis.**
4. On the other hand, heterotrophic algae species can **utilize small organic carbon compounds** that are turned into protein, and oils.
5. Ethanol production from algal starch is like conversion processes of **starch or sugars to ethanol (plant source).**
6. Further, CO₂ produced in industrial flue gases can be used to produce algal biomass.
7. Waste water from industrial and domestic sewage can also be used for the cultivation of algal biomass.
8. Cultivation of microalgae through open ponds is economical but has inherent disadvantages such as low productivity, water loss, low CO₂ utilization, and high affinity to be contaminated by other algal strains.
9. The disadvantages of open ponds led to development of closed photobioreactors, which facilitate higher productivity, less contamination, and less water loss.

9. Annual energy requirement of an industry is 20000 kWh. Calculate the size of wind turbine that is required to be installed to meet the energy requirement and also estimate the cost of installation.

- Coefficient of performance – 0.40
- Transmission losses (rotor to generator) = 0.90
- Generator losses = 0.90
- Wind speed at 15 metre height is 5 metre/sec
- Density of air – 1 kg/m³
- Capacity factor – 0.30
- Number of hours in a year 8760 hours.

Total power density of wind (power per unit area)

$$= \frac{1}{2} \times \text{air density} \times (\text{velocity})^3$$

$$= 0.5 \times 1 \times (5 \times 5 \times 5)$$

$$= 62.5 \text{ watt/m.}$$

$$\begin{aligned} \text{Overall loss factor} &= C_p \times \text{transmission loss} \times \text{generator loss} = 0.40 \times 0.90 \times 0.90 \\ &= 0.324 \end{aligned}$$

Actual power density = Total power density x overall loss factor

$$= 62.5 \times 0.324$$

$$= 20.25 \text{ W/m}^2$$

$$\begin{aligned} \text{Annual energy density (useful)} &= \text{Actual power density} \times \text{number of hours per year} \\ &= 20.25 \times 8760 = 177.39 \text{ kWh/m}^2 \end{aligned}$$

The rotor size and turbine power rating estimation:

Area to be swept by the rotor = Total annual energy required / Useful energy density

$$= 20000 / 177.39$$

$$= 112.75 \text{ m}^2$$

Radius of the rotor blade, (R)

$$\pi R^2 = 112.75$$

$$R = 5.91 \text{ metre.}$$

Power rating of the turbine = Actual power density $\times$ area of the rotor

$$= 20.25 \times 112.75$$

$$= 2.28 \text{ kW}$$

Actual rated power of the turbine rating should be = power rating / capacity factor

$$= 2.28/0.30$$

$$= 7.6 \text{ kW} \sim 8 \text{ kW}$$

Thus, an 8 kW rated power wind turbine will provide desired annual energy under given situation. The cost of a wind turbine of per kW basis runs between Rs.30000 and Rs.50000. For the above calculation a 8 kW wind turbine would cost about $8 \times 40000 = \text{Rs.}320000$.

10. Does Biogas contribute to clean energy? Discuss

Yes, **biogas contributes significantly to clean energy.**

1. **Renewable Resource** – Produced from continuously available organic waste, making it sustainable.
2. **Methane Capture** – Prevents methane (a potent greenhouse gas) from escaping into the atmosphere.
3. **Reduces Fossil Fuel Use** – Substitutes LPG, diesel, and coal for cooking, heating, and power.
4. **Low Emissions** – Burns cleaner than fossil fuels, producing fewer air pollutants and carbon emissions.
5. **Waste Reduction** – Converts organic waste into energy, reducing landfill and environmental pollution.
6. **Produces Organic Fertilizer** – Digestate can replace chemical fertilizers, lowering pollution from agriculture.
7. **Energy Access** – Provides reliable, local energy in off-grid and rural areas.
8. **Improves Indoor Air Quality** – Replaces smoky fuels like firewood, reducing health risks.
9. **Supports Circular Economy** – Turns waste into energy and fertilizer, closing the resource loop.
10. **Encourages Sustainable Agriculture** – Uses animal waste efficiently, promoting eco-friendly farming.

11. Explain the benefits of biogas and Biomass feed stock

1. Renewable and Sustainable Energy

- Both biogas and biomass are derived from organic materials that can be replenished, making them **sustainable alternatives to fossil fuels**.

2. Waste Management

- Utilizes agricultural waste, animal manure, food waste, and municipal organic waste, **reducing landfill use** and environmental pollution.

3. Energy Independence

- Reduces reliance on imported fossil fuels, helping rural and remote areas to become **energy self-sufficient**.

4. Environmental Benefits

- **Lower greenhouse gas emissions** compared to coal or diesel.

5. Rural Economic Development

- Creates **employment opportunities** in rural areas for collecting, processing, and managing feedstock.
- Encourages entrepreneurship in the renewable energy sector.

6. Soil Health and Fertilizer

- Biogas digestate (residue) is a **nutrient-rich organic fertilizer**, improving soil fertility and reducing chemical fertilizer use.

7. Versatility

- Biomass can be converted into electricity, heat, or liquid fuels.
- Biogas can be used for cooking, heating, electricity generation, or as a vehicle fuel (bio-CNG).

8. Energy Storage and Base Load Supply

- Unlike solar or wind, biomass and biogas can provide **continuous, controllable power**, making them reliable base-load energy sources.

9. Reduces Deforestation

- Replaces firewood in rural areas, protecting forests and preserving biodiversity.

12. Government would like to make huge invest on biogas plants across the state. As a renewable pathway student let them know the advantage this creates.

Advantages of Biogas Plant:

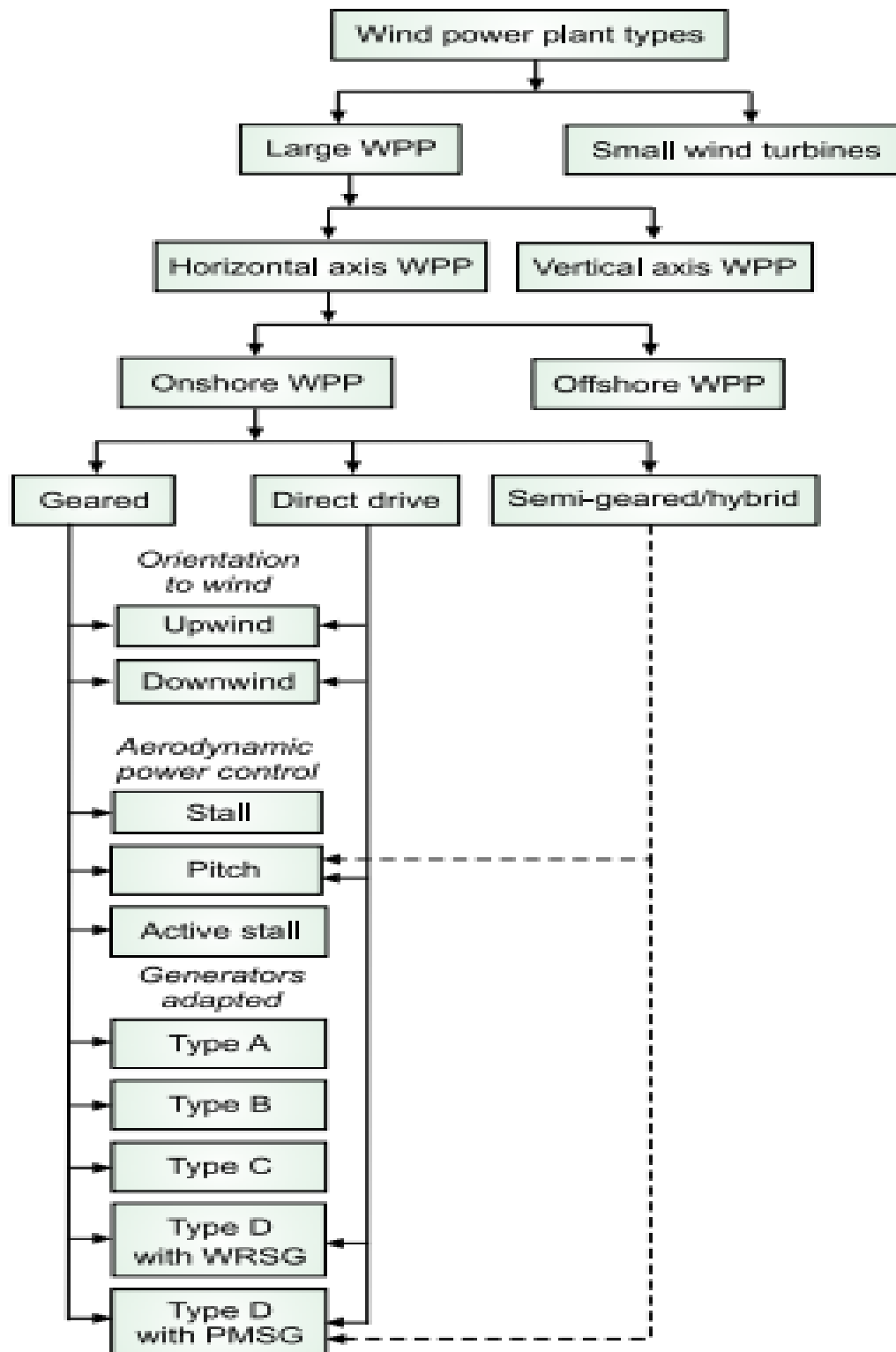
1. **Renewable Energy Source:** Biogas is produced from organic waste, making it a sustainable and eco-friendly energy source.
2. **Waste Management:** It helps reduce the volume of organic waste, minimizing landfill use and environmental pollution.
3. **Reduces Greenhouse Gases:** Captures methane that would otherwise be released into the atmosphere, reducing global warming potential.
4. **Produces Organic Fertilizer:** The by-product (digestate) can be used as a nutrient-rich natural fertilizer, improving soil health.
5. **Low Operational Cost:** After installation, biogas plants are relatively cheap to operate and maintain.
6. **Energy Access:** Provides reliable energy for cooking, heating, and electricity, especially in rural or off-grid areas.
7. **Improves Sanitation:** Enhances hygiene by safely processing animal waste and sewage.
8. **Reduces Dependency on Fossil Fuels:** Offers a clean alternative to coal, LPG, and firewood.

13. Write a note on Wind turbine classes

Wind turbine classes

- Most large wind turbines are certified to IEC standards.
- Class I, II, and III specify the wind speeds for a specific turbine product. Manufacturers who are certifying WPPs, design them based on one of these classes.
- Class I wind turbines are designed to operate in the harshest climates.
- Class II wind turbines are designed for the medium resource sites.
- Class III turbines are designed for low wind resource sites.
- Typically, Class II and III wind turbines have a larger turbine rotor (longer blades) to capture more of the wind energy in lower wind speeds.

14. Discuss classifications of wind power plant.



15. Derive the expression for Energy in the wind

when disc of air of mass m flowing at a speed v has kinetic energy E , then this mechanical energy is converted to electrical power, i.e., energy produced per unit time is:

$$P = \frac{\text{Work}}{\text{Time}} = \frac{\text{Kinetic energy } (E)}{t} \quad (2.4)$$

Kinetic energy E contained in wind $P_{\text{kin}} = 1/2 mv^2$
when,

v = Wind speed (metre per second)

ρ = Density of air (ρ) which is 1.225 kg/m^3 at 15°C at sea level

$m = \rho A d$ (Density $\rho \times$ Area $A \times$ Distance d)
= $(\text{kg/m}^3) \times (\text{m}^2) \times (\text{m}) = \text{kg}$.

The air density also affects the power produced. The air density varies as a function of pressure and temperature in compliance with perfect gas law. Since pressure and temperature vary according to the altitude of the installation site, their combination affects the air density which can be derived from the simplified relation (valid up to 6000 m altitude) given below:

$$\rho = \rho_0 - 1.194 \times 10^{-4} \times H \quad (2.5)$$

where,

ρ_0 = Standard density (1.225) at sea level

H = Height of the installation site above sea level (m).

But when a volume of air passes through an area A per unit time, d becomes 1, then,

$$m = \rho A$$

However, when mass flow rate of air through A is with velocity v , then

$$m = \rho A v \quad (2.6)$$

Therefore, total power contained in the wind flowing through the swept area A will be:

$$\begin{aligned} P_{\text{total}} &= 1/2 (\text{Mass flow rate}) v^2 \\ &= 1/2 (\rho A v) v^2 \\ P_{\text{total}} &= 1/2 (\rho A v^3) \end{aligned} \quad (2.7)$$

$$\text{Unit of } P = (\text{kg/m}^3) \cdot (\text{m}^2) \cdot (\text{m}^3/\text{s}^3) = \text{kg m/s}^2 \text{ m/s} = \text{N m/s} = \text{Watt}$$

16. What are the factors that depends on the Wind Power plant performance?

The performance of a WPP also depends on the following wind speed values:

- **Start-up speed:** With this wind speed, the rotor just starts to rotate and the electric generator generates a voltage which increases when the wind speed rises.
- **Cut-in speed:** At this wind speed, the voltage which is high enough for the WPPs to be connected to the grid. It ranges between 2 m/s–4 m/s.
- **Rated speed:** It is the wind speed at which the rated power is reached and its range is between 10 m/s–14 m/s.
- **Cut-off speed:** It is the wind speed beyond which the rotor is stopped to save the WPP from possible damage. It ranges between 20 m/s–25 m/s.

- **Survival speed:** It is the wind speed which the WPP withstands without getting damaged in the worst storm which may occur on the installation site during the design lifetime (50-year recurrence period). It ranges between 55 m/s–65 m/s.

17. Discuss from the data given in the table

Crop	yield per hectare	%Efficiency
Maize	30000 tonnes	75
Rice husk	50000 tonnes	76
Sunflower	10000 tonnes	60

Which crop provides the suited for biogas production discuss.

Step 1: Calculate Effective Biogas Yield

To compare fairly, we calculate **effective biogas yield** using:

Effective Yield = Yield × (Efficiency ÷ 100)

- **Maize:**
 $30,000 \times (75 \div 100) = 22,500 \text{ tonnes}$
- **Rice Husk:**
 $50,000 \times (76 \div 100) = 38,000 \text{ tonnes}$
- **Sunflower:**
 $10,000 \times (60 \div 100) = 6,000 \text{ tonnes}$

Step 2: Discussion

- **Rice husk** gives the **highest effective biogas yield** at **38,000 tonnes per hectare**, due to:
 - The **highest raw yield** (50,000 tonnes), and
 - The **highest efficiency** (76%).
- **Maize** comes second with 22,500 tonnes. It has moderate yield and good efficiency.
- **Sunflower** has the **lowest effective yield**, both due to low production and lower efficiency.

Conclusion

Rice husk is best suited for biogas production among the three crops.

It provides the highest combination of **biomass yield** and **conversion efficiency**, making it the most efficient and productive option for biogas generation.

18. Discuss environmental impacts on producing biogas?

Pros (Positive Environmental Impacts)

1. Reduces Greenhouse Gases

- Captures methane that would otherwise escape into the atmosphere, helping fight climate change.

2. Improves Waste Management

- Converts organic waste (manure, food waste, sewage) into energy, reducing landfill use and pollution.

3. Produces Renewable Energy

- Biogas is a clean, sustainable alternative to fossil fuels, lowering carbon emissions.

4. Reduces Deforestation

- Replaces firewood in rural areas, helping preserve forests and biodiversity.

5. Generates Organic Fertilizer

- The digestate by-product can be used as a natural fertilizer, reducing the need for chemical fertilizers and their associated pollution.

Cons (Negative Environmental Impacts)

1. Methane Leakage Risk

- Poorly maintained systems can leak methane, a potent greenhouse gas.

2. Water Contamination

- Improper handling of digestate can pollute surface and groundwater.

3. Air Emissions from Combustion

- Burning biogas still releases CO₂ and small amounts of nitrogen oxides (NO_x).

4. Odor Issues

- If feedstock is not managed well, biogas plants can emit foul odors.

5. Land Use Concerns

- Large-scale biogas production using energy crops may compete with food production and impact ecosystems.

19. Differentiate between type A and type B wind turbine generators?

Wind turbine generators are classified into types (A to D) based on their **generator type** and **grid connection method**. Here's a comparison of **Type A** and **Type B** wind turbines:

- **Type A:** Fixed-speed, simple, low-cost design using **squirrel-cage induction generators**. Less efficient, higher stress under wind fluctuations.
- **Type B:** Allows **limited variable speed** by controlling rotor resistance in **wound-rotor induction generators**. Offers better performance and lower mechanical stress than Type A but at a slightly higher cost and complexity.

Feature	Type A Wind Turbine Generator	Type B Wind Turbine Generator
Generator Type	Squirrel-cage induction generator	Wound rotor induction generator
Rotor Speed Control	Fixed-speed (very limited speed variation)	Limited variable speed (slip control using external resistor)
Slip Power Control	Not possible	Possible via external variable resistors
Power Electronics Usage	None or minimal (usually only for control systems)	Minimal (power electronics used for rotor resistance control)
Efficiency	Lower, due to fixed speed operation	Slightly higher, due to ability to adjust rotor resistance
Mechanical Stress	Higher (fixed speed causes more stress during wind gusts)	Lower (slip control reduces torque fluctuations)
Cost	Lower (simple and robust design)	Slightly higher (due to external rotor circuit)
Grid Connection	Directly connected to the grid	Directly connected to the grid with controlled rotor circuit
Control Complexity	Very simple	Moderate complexity

20. What type of waste produces Biogas? Discuss

wide variety of waste material breaks down into biogas, including **animal manure, municipal rubbish or waste, plant material, food waste or sewage.**

Waste Type	Example Sources	Biogas Potential
Animal Waste	Cow dung, pig manure, poultry litter	High
Agricultural Waste	Crop residues, spoiled grains	Medium to High
Food Waste	Kitchen scraps, hotel/restaurant waste	Very High
Sewage Sludge	Municipal wastewater plants	Medium
Industrial Organic Waste	Breweries, dairy, slaughterhouse waste	High
Municipal Organic Waste	Biodegradable garbage	Medium
Green Waste	Grass, leaves, garden trimmings	Low to Medium

21. write any 5 difference between horizontal and vertical axis wind turbines?

Feature	Horizontal-Axis Wind Turbine (HAWT)	Vertical-Axis Wind Turbine (VAWT)
Axis Orientation	Rotor shaft is horizontal to the ground.	Rotor shaft is vertical to the ground.
Wind Direction	Requires the turbine to face into the wind.	Can capture wind from any direction.
Efficiency	Higher efficiency due to better wind capture at higher altitudes.	Lower efficiency and wind capture at lower heights.
Power Output	Typically produces greater power output.	Generally produces lower power output.
Maintenance	More complex due to components like gearboxes and yaw mechanisms.	Simpler maintenance with fewer moving parts.
Installation Location	Suited for open, vast areas or offshore locations with steady winds.	Can be installed in urban areas or locations with turbulent winds.
Noise and Vibration	Can produce more noise and vibration due to high-speed operation.	Typically quieter with less vibration.
Aesthetics	Often larger and considered an eyesore in some areas.	Smaller , often more aesthetically acceptable, especially in residential areas.

Unit 5

1. Can we store hydrogen in any form? Justify your answer.

Hydrogen can be stored physically as either a gas or a liquid. Storage of hydrogen as a gas typically requires high-pressure tanks (350–700 bar [5,000–10,000 psi] tank pressure). Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen at one atmosphere pressure is -252.8°C .

Hydrogen gas is highly flammable and can easily escape containment. If hydrogen gas escapes containment, it can corrode metals. This, in turn, can make these contaminated metals brittle and prone to breaking.

Since hydrogen is a very light gas, its storage is an essential obstacle to overcome. Technical means are required to store it at low temperature or high pressure.

2. Name the device or process used to produce hydrogen. How will you produce hydrogen in cost effective manner.

Electrolysis is a promising option for carbon-free hydrogen production from renewable and nuclear resources. Electrolysis is the process of using electricity to split water into hydrogen and oxygen. This reaction takes place in a unit called an electrolyzer.

The **carbon monoxide is reacted with water to produce additional hydrogen**. This method is the cheapest, most efficient, and most common. Natural gas reforming using steam accounts for the majority of hydrogen produced in the United States annually.

3. Mr X has purchased a land in a remote village which has no access to electricity to build a energy efficient sustainable farm house. His idea is to Combine farming with PV, What idea he has and discuss advantages and disadvantages of his idea

Mr. X's idea of combining farming with **PV (Photovoltaic)** refers to the concept of **agrivoltaics** — a system where **solar panels are installed over agricultural land** to simultaneously produce solar energy and grow crops. This concept allows for dual use of land for **energy generation** and **agriculture**, which is especially useful in remote locations without access to grid electricity.

Advantages of Mr. X's Idea

1. Clean Energy Generation

- Solar PV provides a renewable, non-polluting source of energy — ideal for remote areas with no grid access.

2. Land Efficiency

- Agrivoltaics enables dual use of land for farming and power generation, maximizing utility.

3. Improved Crop Conditions

- Solar panels can provide shade, reducing heat stress on crops and lowering evaporation rates, which is useful in hot climates.

4. Lower Operational Costs

- Reduces or eliminates reliance on diesel generators or other costly energy sources.

5. Sustainable Living

- Supports eco-friendly and energy-independent rural development, aligning with green building and sustainability goals.

Disadvantages and Challenges

1. High Initial Cost

- Installing solar panels and battery storage requires significant upfront investment.

2. Technical Maintenance

- Solar systems require occasional cleaning, maintenance, and technical knowledge which might not be readily available in remote villages.

3. Shading Impact

- Not all crops benefit from partial shading; crop selection must be compatible with reduced sunlight.

4. Infrastructure Requirements

- Battery systems, inverters, and water pumps must be planned carefully for efficient integration.

5. Local Acceptance

- Villagers unfamiliar with agrivoltaics might resist or misunderstand its benefits.

4. Analyse the grid applications of battery energy storage system.

Grid applications of battery energy storage system

- Electricity can be purchased and stored during off-load period when the cost is low and used or sold during peak load period for higher rate.
- Reduces the need to buy new generator to meet additional load demand.
- Helps to maintain grid frequency.
- Helps to maintain reserve capacity which is used at the time of grid failure.
- Used in black start restoration, to start the generators when there is total failure of power in the generating station.
- Helps to maintain good power quality by maintaining the frequency, voltage and power factor at required value.

5. Discuss the concept of micro grid and its benefits.

Micro grids are local power grids that can be operated independently. It can be used to power a single building, like a hospital or police station, or a collection of buildings, an industrial park, university campus, military base or neighborhood.

Groups of microgrids that are linked together can also power bigger areas, like towns or cities.

Benefits: -

- When storms or power outages shut down the main electricity grid in an area, large numbers of homes, businesses and critical services can be affected.
- Micro grids can switch away from the main grid and continue to provide power during emergencies like these. This process is known as ‘islanding’.
- Micro grids can also provide power in remote places that have no access to electricity.
- As they can generate and store their own energy, micro grids increasingly use renewable energy – like solar panels, wind turbines and batteries. This means more micro grids would help reduce greenhouse gas emissions.

6. **Discuss Battery Energy Storage Systems (BESS) and its components.**

A BESS is a type of energy storage system that uses batteries to store and distribute energy in the form of electricity. These systems are commonly used in electricity grids and in other applications such as electric vehicles, solar power installations, and smart homes.

At its most basic level, a BESS consists of one or more batteries that store electrical energy for use at a later time. This stored energy can then be drawn upon when needed to meet various demands for power across different applications.

BESS can also provide advantages over other energy storage systems, including greater efficiency and flexibility, faster response times when powering equipment or devices, and lower costs overall.

BESS and its components

Battery Energy Storage Systems offers more than just a standard battery. It is fully packed with technologies allowing its system to capture charge and execute discharge. The following are the typical technologies it includes:

Inverters

Inverters are devices that transform direct current (DC) to alternating current (AC). AC is the type of electricity used in homes and businesses.

Control Components

The control components of a BESS manage the charging and discharging of the batteries and regulate the flow of electricity to and from the grid.

Integrated Sensors

Integrated sensors monitor the BESS's performance and conditions, providing valuable data to help optimize its operation.

Multiply Battery Modules

Multiple battery modules are composed of multiple batteries that work together to store and release energy.

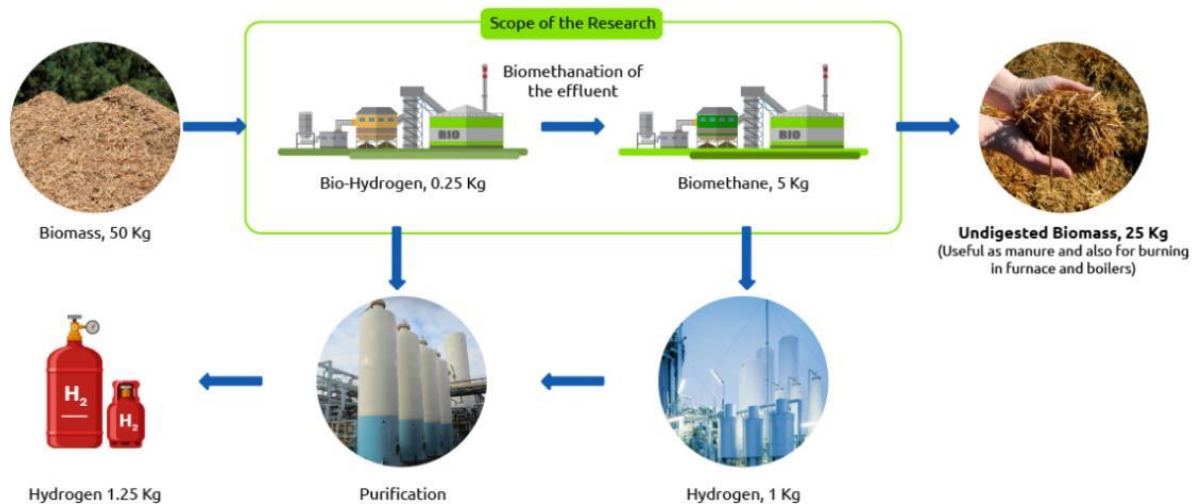
7. How environmentally friendly and sustainable is hydrogen fuel cell technology?

1. The **exhaust gas** from a hydrogen engine consist of pure water vapor. Hydrogen fuel cell technology is therefore locally emission-free.
2. Hydrogen production requires electrical energy to break water down into its constituent elements, hydrogen and oxygen, via the process of electrolysis. If the electricity used comes from renewable energy sources, the hydrogen production has a neutral **carbon footprint**. If, on the other hand, fossil fuels are used, this will ultimately have a knock-on effect on the carbon footprint of the fuel cell cars using the hydrogen.
3. One disadvantage of producing hydrogen is the losses during electrolysis.
4. The **overall efficiency** in the “power to vehicle drive” energy chain is therefore only half the level of a BEV.
5. Hydrogen is also a by-product of many industrial processes, where all too often it is treated as waste with no further use. The fuel cell battery offers a way to **upcycle** this hydrogen, although it must be cleaned first.
6. The **energy balance sheet** for hydrogen fuel cell cars also has to include the transportation and storage of the hydrogen. Depending on the transportation technology used (liquid or gaseous), different costs for compression, cooling, transport and storage arise.
7. Due to its better transportability and storage ability, the trend is towards liquid hydrogen. Nevertheless, the transportation and storage of hydrogen are – at this stage – still a good deal more complex and energy-intensive than for gasoline or diesel.
8. In contrast to fossil fuels, hydrogen can be produced anywhere there is access to electricity and water, theoretically even at the actual filling stations for fuel cell cars.
9. A more highly developed infrastructure could thus shorten transportation distances significantly in future.

8. A company is planning to generate hydrogen using biomass, suggest the technology and explain with a block diagram.

We can suggest Hydrogen Generation by Anaerobic Digestion of Biomass.

Agricultural residues such as rice straw, wheat straw, sugarcane tops, Sugar industry waste such as press mud and bagasse, food industry waste and dedicatedly cultivated biomass are used as input feed.



9. Differentiate between Green hydrogen and black hydrogen?

Hydrogen itself is a **color less** gas but there are around nine color codes to identify hydrogen. The colors codes of hydrogen refer to **the source or the process used to make hydrogen**. These codes are: green, blue, grey, brown or black, turquoise, purple, pink, red and white.

- **Green hydrogen** is produced through **water electrolysis** process by employing renewable electricity. The reason it is called green is that there is no CO₂ emission during the production process. Water electrolysis is a process which uses electricity to decompose water into hydrogen gas and oxygen.
- **Black or brown hydrogen** is produced from **coal**. The black and brown colours refer to the type bituminous (black) and lignite (brown) coal. The **gasification of coal** is a method used to produce hydrogen. However, it is a very polluting process, and **CO₂ and carbon monoxide** are produced as by-products and released to the atmosphere.

10. Differentiate between Polymer Electrolyte Membrane and alkaline Electrolysers.

Polymer Electrolyte Membrane Electrolysers

Electrolyte is a polymer material.

Anode Reaction: $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

The electrons flow through an external circuit and Hydrogen ions move across the electrolyte to the cathode.

Cathode Reaction: $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2$

At the cathode, hydrogen ions combine with electrons from the external circuit to form hydrogen gas

Alkaline Electrolysers

Electrolyte :- Liquid alkaline solution of sodium or potassium hydroxide.

Potassium hydroxide splits into (K^+) and (OH^-) .

Anode :- Hydroxide ions (OH^-) move to the anode liberating oxygen. Electrons generated moves to cathode.

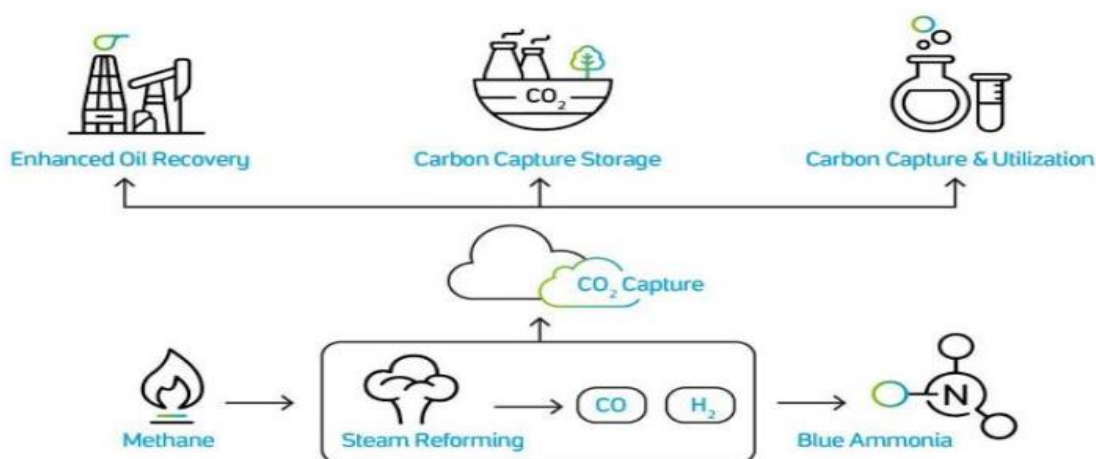
$2\text{OH}^- \rightarrow \frac{1}{2}\text{O}_2 + \text{H}^+ + 2\text{e}^-$

Cathode :-Water molecules reacts with electrons to generate hydrogen

$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^- + \text{H}_2$

Overall $\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2}\text{O}_2$

11. Discuss the method of producing blue hydrogen



Blue hydrogen is hydrogen that is produced from natural gas (methane) using processes like Steam Methane Reforming (SMR) and combined with Carbon Capture and Storage (CCS). The goal is to produce hydrogen while capturing and storing the CO₂ emissions, making it a lower-carbon alternative to grey hydrogen.

Depending on the nature of the method of its extraction, hydrogen is categorised into three categories, namely, Grey, Blue and Green.

The CO₂ generated during this process is captured and stored underground, preventing its release into the atmosphere.

Hence, blue hydrogen is a lower-emission alternative to natural gas and other fossil fuels.

12. Discuss the concept of gravity storage of electricity.

Gravity storage of electricity is a method of storing energy by using gravitational potential energy. It involves lifting a heavy object (like a mass or water) to a height using electricity when supply is abundant, and then releasing it to generate electricity when needed. This concept is a form of mechanical energy storage and is being explored as a sustainable, long-duration energy storage solution.

Basic Principle

Input (Charging): Electricity is used to lift a heavy mass or pump water to a higher elevation.

Storage: Energy is stored in the form of gravitational potential energy.

Output (Discharging): The mass is allowed to fall or water flows down, driving a generator to produce electricity.

Types of Gravity Storage Systems

1. Pumped Hydro Storage (PHS) (Most Common)
2. Solid Mass Gravity Systems
3. Underground Gravity Systems

13. List the Advantages and disadvantages of Gravity Storage of electricity.

Advantages of Gravity Storage

- Long Lifespan: Mechanical systems can last for decades with low degradation.
- Environmentally Friendly: No chemical waste or harmful materials.
- Fast Response Time: Quick to store and release power.
- Suitable for Renewable Integration: Ideal for balancing intermittent sources like solar and wind.
- Location Flexibility (especially for non-water systems): Can be built in places without elevation or water access.

Disadvantages of Gravity Storage

- High Initial Costs: Especially for infrastructure (cranes, shafts, reservoirs).
- Location Constraints: Pumped hydro requires suitable geography (elevation and water supply).
- Lower Energy Density: Compared to batteries, more space and mass are needed to store the same energy.
- Complex Engineering: Especially for deep shaft or tower-based systems.

14. Write a note on Present trends of hydrogen Technology in India

India is making significant strides in hydrogen technology, positioning itself as a global leader in green hydrogen production and utilization. The National Green Hydrogen Mission (NGHM), launched by the Government of India, aims to establish a robust ecosystem for green hydrogen, targeting the production of 5 million tonnes annually by 2030.

Key Developments in India's Hydrogen Technology Landscape

1. National Green Hydrogen Mission (NGHM)

- **Objective:** To make India a global hub for the production, usage, and export of green hydrogen and its derivatives.
- **Targets:** Produce 5 million tonnes of green hydrogen annually by 2030.
- **Infrastructure:** Development of green hydrogen hubs with an outlay of ₹400 crore up to 2025–26 .

2. Industrial Initiatives and Collaborations

- **Indian Oil Corporation Ltd (IOCL):** Awarded a tender to L&T Energy Green Tech to construct a green hydrogen production facility at its Panipat refinery, aiming to produce 10,000 tonnes annually.
- **Coal India Limited (CIL):** Signed an agreement with AM Green to supply 4,500 MW of renewable energy for green ammonia production, integrating solar, wind, and pumped hydro storage.
- **Bharat Petroleum Corporation Limited (BPCL):** Entered a joint venture with Sembcorp to develop green hydrogen and renewable energy projects across India, including initiatives in green ammonia production and port emissions reduction.

3. Advancements in Hydrogen Mobility

- **Chhattisgarh:** Launched India's first hydrogen-powered fuel cell truck to support cleaner transportation and reduce reliance on diesel, marking a significant step toward sustainable transportation .
- **Pilot Projects:** India initiated pilot tests of hydrogen-fueled commercial vehicles in 2025, aligning with plans for hydrogen truck production to begin in the last quarter of 2026.

4. Policy and Regulatory Support

- **Hydrogen Purchase Obligations (HPOs):** Domestic hydrogen manufacturers are advocating for HPOs to ensure a consistent market for hydrogen, particularly targeting refineries and ammonia plants, to prevent the stranding of investments in hydrogen production facilities.

5. Technological Innovations

- **Electrolyzer Technologies:** Advancements in proton exchange membrane (PEM) and solid oxide electrolyzers (SOE) are enhancing efficiency and reducing costs, contributing to the scalability of green hydrogen production.
- **Storage Solutions:** Innovations in hydrogen storage and distribution, including solid-state hydrogen storage using metal hydrides or chemical compounds, are becoming viable solutions.

15. Is a hydrogen powered cars the same as an hybrid Vehicle? Justify the Answer.

No, a **hydrogen-powered car is not the same as a hybrid vehicle**. Although both aim to reduce environmental impact and fossil fuel use, they operate using **different technologies** and **energy sources**.

1. Hydrogen-Powered Car (Fuel Cell Electric Vehicle – FCEV)

- **Energy Source:** Uses **hydrogen gas**, stored in tanks.
- **Working Principle:** A **fuel cell** combines hydrogen with oxygen from the air to produce **electricity**, which powers an **electric motor**.
- **Emissions:** Only **water vapor** is emitted — **zero tailpipe emissions**.
- **Example:** Toyota Mirai, Hyundai NEXO.

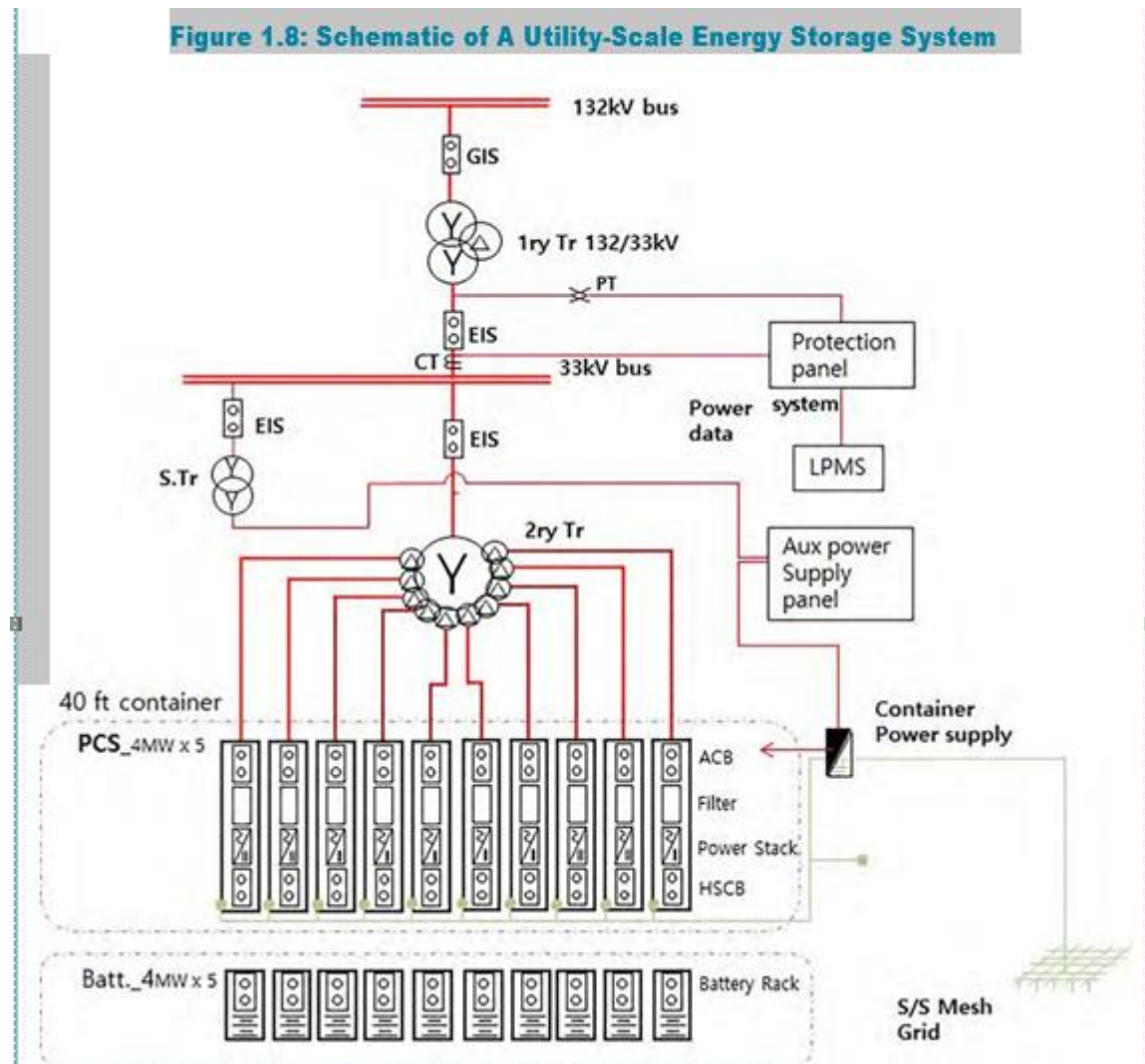
2. Hybrid Vehicle

- **Energy Source:** Combines a **petrol/diesel engine** with an **electric motor and battery**.
- **Working Principle:** The vehicle switches between or combines the internal combustion engine and electric motor to optimize fuel efficiency.
- **Emissions:** Emits CO₂ and other pollutants when running on fuel.
- **Types:**
 1. **Mild Hybrid:** Electric motor assists engine but can't run the car alone.
 2. **Full Hybrid:** Can run on electric motor alone for short distances.
 3. **Plug-in Hybrid (PHEV):** Has a larger battery that can be charged and used for longer electric-only trips.

Key Differences

Feature	Hydrogen-Powered Car (FCEV)	Hybrid Vehicle
Fuel Type	Hydrogen	Petrol/Diesel + Electricity
Emission Output	Zero (only water vapor)	Reduced, but not zero
Powertrain	Electric motor powered by fuel cell	Combines engine and electric motor
Refueling/Recharging Time	~5 minutes (hydrogen tank)	Fast refueling + some recharge time
Environmental Impact	Very low	Moderate
Infrastructure	Requires hydrogen stations	Widely supported with gas stations

16. Draw the SLD of utility scale energy storage system and explain.



A utility-scale energy storage system (ESS) is a large-scale energy storage solution used by power utilities to store and release electricity to the grid. These systems help balance electricity supply and demand, integrate renewable energy sources, and enhance grid reliability and stability.

Utility-scale ESS stores **excess electricity** (often from renewable sources like solar or wind) when demand is low and releases it when demand is high or during power shortages. These systems are **grid-connected** and are designed to operate at **megawatt (MW) to gigawatt (GW)** scale.

Types of Utility-Scale Energy Storage Technologies

1. Battery Energy Storage Systems (BESS)
2. Pumped Hydro Storage
3. Compressed Air Energy Storage (CAES)
4. Gravity-Based Storage
5. Flywheel Energy Storage

Advantages of Utility-Scale Energy Storage

1. **Grid Stabilization:** Helps maintain voltage and frequency stability.
2. **Renewable Integration:** Stores excess solar/wind energy for later use.
3. **Peak Shaving:** Reduces the need for expensive peaking power plants.
4. **Backup Power:** Provides electricity during outages or grid failures.
5. **Energy Arbitrage:** Buy electricity when it's cheap, sell when it's expensive.

Disadvantages of Utility-Scale Energy Storage

1. **High Initial Cost:** Especially for large battery or hydro systems.
2. **Land and Resource Needs:** Pumped hydro and CAES require specific geographic conditions.
3. **Limited Duration:** Battery systems typically store energy for a few hours.
4. **Recycling and Lifecycle Issues:** Especially with batteries (chemical waste, rare metals).